EURASIP Journal on Information Security

RESEARCH

Open Access

An IoT system for access control using blockchain and message queuing system

Nilima Karankar[1*] and Anita Seth[2]

Abstract

In the Internet of Things (IoT) security, traditional access control methods are increasingly insufficient to address the complexities and scalability challenges posed by vast networks of interconnected devices. This paper introduces an advanced IoT system for access control that leverages blockchain methodology and a message queuing system to enhance security, clarity, and efficiency. The suggested framework integrates blockchain to provide a decentralized and unchangeable record keeper for access control transactions, ensuring that access rights and logs are absolute. In line with this, we employ a message queuing system to facilitate real-time communication and coordination among IoT gadgets, access control nodes, and users. This combination allows for secure, scalable, and efficient management of access requests and permissions across a dynamic IoT environment. This paper details the design of the system, the integration of blockchain, and message queuing components and evaluates the system's performance through a sequence of modeling and real-world scenarios. The outcome shows significant improvements in security, transparency, and operational efficiency compared to conventional access control mechanisms. This innovative approach paves the way for more robust and reliable IoT access control solutions, addressing key challenges in modern IoT deployments like scalability.

Keywords Internet of Things (IoT), Blockchain, Security, Scalability, Message queuing, Transparency, Operational efficiency

1 Introduction

Internet of Things (IoT) devices are increasingly both widespread and greatly influencing different aspects of our daily lives. IoT technologies provide many other conveniences in several areas, such as call fields, smart homes, transportation, and voting. The IoT has grown significantly, with these devices generating large amounts of data that need to be communicated over networks for real-time data processing [1]. Despite the myriad data being generated, collection and efficient storage of such large volumes of data remain a daunting task due to the resource constraints of IoT devices. Securing and scaling IoT systems are still a fundamental concern. The network transfers important IoT data, which may contain sensitive information. For example, if a patient is in a severe situation, his body is connected with the IoT devices and generates data, which is transmitted to the doctor's end so that the doctor can monitor the patient from anywhere. Such data is a very sensitive information and may generate emergency alerts. Securing these gadgets and such a type of network requires a strong access control mechanism. The access control mechanism [2] instructs the system to give access or not. Access control systems in IoT are capable of efficiently avoiding a set of connection violations, for example, illegitimate intrusion, and know how to stay away from the outflow of sensitive data. Classical access control systems include discretionary access control (DAC), mandatory access control

*Correspondence:
Nilima Karankar
nkarankar@ietdavv.edu.in
[1] Department of Computer Engineering, Institute of Engineering and Technology (IET), Devi Ahilya Vishwavidyalaya, Indore, India
[2] Department of Electronics and Telecommunications, Institute of Engineering and Technology (IET), Devi Ahilya Vishwavidyalaya, Indore, India

(MAC), and role-based access control (RBAC) [3]. DAC is one of the least restrictive access controls, and it allows multiple administrators to access the system. These multiple administrators must have proper communication between them about access; it may lead to a state. Only one administrator can handle all access rights in MAC, potentially slowing down the system's overall performance. In the case of RBAC, the admin assigns the role to each user and manages access. This system actually establishes many-to-many associations between users and their responsibilities. However, these conventional access control methods rely on centralization and lack transparency, leading to challenges such as a single point of failure and significant privacy concerns. Therefore, we need an access control system that relies on decentralized management. Dispersed record-keeper technology provides transparent, distributed, and secure systems that run on distributed ledgers. This paper proposes a blockchain-based IoT device access control system. Access rules are written in the form of smart contracts (SC), and SC is stored in blockchain. Blockchain is a shared and immutable ledger, which can make data tracking and transaction recording in networks easier. Blockchain provides more security to the IoT and solves access-storage-related issues. So, in this work, a system to control access is proposed to access the IoT system smoothly and to provide security and scalability using blockchain technology and attribute-based access control (ABAC) policy.

The primary contributions of this paper are as follows:

- A scalable and decentralized IoT access control system by combining blockchain and RabbitMQ
- The use of smart contracts for enforcing the fine-grained access control policies formulated with ABAC
- An asynchronous and reliable message queuing layer for microservices to process the high volume of IoT transactions
- Experimental study demonstrating the advances in execution time, load balancing, and system scalability

We proposed an IoT system that utilizes blockchain and ABAC policies to enhance both its safety and scalability. To meet the IoT security requirements using a private blockchain, we developed a lightweight and robust access control system. In our proposed architecture, RabbitMQ (AMQP) is used specifically for handling blockchain operations like managing write and retrieval processes, not for direct IoT communication. RabbitMQ was chosen because it offers reliable message delivery, transaction support, and message persistence, which are crucial for scaling blockchain operation features that MQTT lacks. Additionally, RabbitMQ's security (TLS encryption and access control) aligns with the blockchain's security model. If necessary, we can integrate an MQTT-to-RabbitMQ bridge to connect IoT devices and maintain robust blockchain handling. The architecture, RabbitMQ (which uses AMQP), is employed solely to scale blockchain operations, such as saving and retrieving IoT data. We do not integrate AMQP at the internal blockchain level for security or consensus mechanisms. Instead, its native cryptographic protocols manage blockchain security, ensuring data integrity and access control. RabbitMQ only enhances transaction processing and does not disrupt the blockchain's security framework.

The computational complexity of the proposed scheme is determined by two key factors: blockchain operations and RabbitMQ-based scaling. Blockchain operations, such as storing and retrieving IoT credentials and managing access control, depend on the consensus mechanism used. For permissioned blockchains, where transactions are validated by a fixed set of nodes, the complexity is typically O(log n) to O(n), depending on the data structure used for storage and retrieval. In contrast, public blockchains with proof-of-work mechanisms may have a higher complexity due to cryptographic computations. To enhance scalability, our approach integrates RabbitMQ for message queuing, which efficiently asynchronously manages transaction processing, reducing blockchain congestion. The complexity of message queuing and dispatching in RabbitMQ is approximately O(1) to O(log n), ensuring minimal additional overhead while significantly improving system throughput. This hybrid approach optimizes performance, balancing computational efficiency and scalability for secure IoT data management.

Blockchain operations, RabbitMQ-based message queuing, IoT data transactions, and the use of electricity for computational resources impact the power consumption of the proposed scheme. Traditional blockchain systems have high-energy demands due to frequent consensus operations, cryptographic computations, and data replication across nodes. Our approach optimizes power consumption by offloading transaction processing to RabbitMQ, which operates with significantly lower computational overhead and reduces direct blockchain interactions. This method leads to lower CPU usage, less strain on computational resources, and reduced overall electricity consumption. The use of permissioned blockchains in our model further improves energy efficiency, as they rely on consensus mechanisms like PBFT or RAFT, which do not require intensive mining processes.

Our system can be deployed on low-power edge devices or cloud-based infrastructure, ensuring an optimized balance between security, scalability, and power efficiency. Network impairments, such as latency, packet loss, and network congestion, can impact the performance of both blockchain transactions and RabbitMQ message processing. RabbitMQ's asynchronous messaging and fault-tolerant architecture help mitigate delays by ensuring message delivery even under unstable network conditions. Since blockchain operations are decentralized, temporary network disruptions do not compromise data integrity but may cause delays in transaction finalization.

We have strengthened our security analysis by explicitly defining security objectives based on the CIA (confidentiality, integrity, availability) triad. We now provide specific attack scenarios and potential vulnerabilities in traditional access control systems and explain how our proposed blockchain-integrated ABAC system mitigates them. We have also elaborated on how blockchain's immutability, transparency, and decentralized nature contribute to enhancing security in IoT access control. We now clearly outline the properties of blockchain and ABAC technologies that enhance security, access control flexibility, and audibility. The uniqueness of the proposed architecture is the message queuing system, which will increase the scalability of an IoT system, and every single transaction/message will be acknowledged. We use token-dependent control for access to ensure that only approved users can use the system, eliminating the need for third-party involvement.

1.1 Motivation

IoT devices are on the rise, and one can rely on access control mechanisms that are secure and scalable. Centralized access control systems can have disadvantages such as security risks, single points of failure, and limited scalability. Blockchain-based decentralized security and RabbitMQ-based queuing for rapid message exchange, and this can be a robust solution for real-time IoT access control.

IoT access control failure can cause the following: Unauthorized access to sensitive IoT devices and data leading to invasion of user privacy and system tampering, data breaches where unauthorized parties can exfiltrate sensitive information from connected devices, and service disruption including the threat of a denial of service (DOS) that can jeopardize the availability of critical IoT infrastructure. These consequences further emphasize the significance of our proposed blockchain-based ABAC for preventing the aforementioned security breaches and thus providing dependable operations of IoT systems, as we stress in the restatement of the problem and the motivation.

The rest of this paper is structured an overview of the literature is included in second section, the proposed architecture and algorithm are shown in the third section, the implementation is shown in the fourth section, results are presented in the fifth section, and, finally, the work is concluded in the last section.

2 Literature survey

An Internet of Things system can be made scalable and safe with a variety of architectures and frameworks that have been proposed. Fabric is an IoT solution to control access depending on the Hyperledger Fabric blockchain architecture with access control (ABAC). Three different kinds of smart contracts are included in the structure: access contract (AC), policy contract (PC), and device contract (DC). Device-generated resource data has a mechanism in DC to query it and to store the Uniform Resource Locator (URL) of the data. Administrators can manage ABAC policies using a PC. The main software used to give regular users access control is called AC. Fabric-IoT, in conjunction with ABAC and blockchain technology, may offer decentralized, fine-grained, and dynamic access control management in IoT. Two sets of sample tests are planned to test the system's performance. The findings show that Fabric-IoT can preserve high throughput in a high-volume request environment while also quickly reaching consensus in a dispersed structure to guarantee data consistency [4].

Author [5] suggested the fabric blockchain data transmission technique for Industrial IoT. It is a dynamic secret sharing mechanism, power data transmission, and data storage optimized. Furthermore, the suggested approach outperforms in terms of share management and decentralization. Power blockchain only works for scale applications. Data security should be authenticated.

The author proposed a blockchain-based storage and sharing scheme for educational records named EduRSS. To guarantee data storage security, out-of-chain documents are regularly attached with hash sequences from the blockchain. To handle record, encryption, and digital signature, cryptography methods are used. To evaluate the efficacy of EduRSS, they created and experienced an evidence-of-concept system. According to the comparative safety study, EduRSS is secure and has an inferior computational cost than the ciphertext policy-attribute-based encryption(CP-ABE) and multi-authority MA-CPABE systems [6]. More functions need to be added into the framework like support for educational records, certification of external institutions, and retrieval of educational records. Decentralized storage technologies such as InterPlanetary File System (IPS) must be used.

The framework ensures that users and IoT gadgets are authorized, and that the services they receive are authenticated as well, thanks to an authentic IT gateway. Additionally, the framework gives users inside the IoT domain tokens to properly access IoT resources. It is developed as a distributed application with a smart contract [7]. The proposed framework is implemented through Node.js. Node.js is single threaded. Security and scalability are a big research challenge.

Due to resource constraints in IoT, blockchain cannot be directly implemented on it. In order to address these challenges, authors proposed a local peer network to bridge the gap [8]. Local Peer (LPeer) network acts as an interface between IoT devices and blockchain network. Local Peer includes certification authority and provides authentication and registration of gadgets/users in network. The proposed architecture improves the scalability of blockchain in terms of ledger size and transaction rate, and due to consensus mechanism of blockchain, LPeer will not allow illegal access. But it focuses only on scalability and resource conservation using existing transaction structure without optimization. It needs to work on unified transaction structure.

In [9], existing blockchain methods are not able to be amplified with respect to vast IoT data with no failure in velocity and competence. To solve this issue, the author used the idea of offline and side chain data storage space. To keep track of all the activities and to ensure data authentication, authorization smart contracts are used. The architecture is very scalable with respect to avg. Latency, throughput, and cost are effectively utilized in an Internet of Things system. Architecture framework is implemented for the healthcare system using blockchain. There is a need to enhance the functionality of the proposed decentralized distributed blockchain application.

This paper is targeting the drawbacks of the single failed and untraceable process of traditional access control. blockchain-based method for IoT control of access is suggested, utilizing the decentralization, transparency, and immutability of blockchain technology. The projected method uses the blockchain to implement control of access policy choices as smart contracts, ensuring the transparency of the entire access authorization process. An off-chain is suggested. Decisions on control of access are implemented through the blockchain's smart contract implementation. The combined effect of the chain (off-chain and on-chain) reduces the storage burden on the blockchain [2].

Existing conventional system of control of access provides safety and organization to the IoT organization. But this mechanism depends on inner ability to manage which may guide to a single point of failure, low scalability, and lack of privacy. So, the author proposed a decentralized control of the access system using blockchain. DAC still suffers from scalability and high computational complexity. Blockchain-based multichain code control of access is proposed to protect the security and privacy of an IoT system. To extend the scalability of IoT system, clustering is used with blockchain managers. Three clusters, edge blockchain manager (EBCM), aggregated edge blockchain manager (AEBCM), and cloud consortium blockchain manager (CCBCM), have different responsibilities. In this paper, permissioned blockchain HLF (Hyperledger Fabric) is used to implement a proof of concept [3].

Secure communication between billions of gadgets is necessary for the IoT ecosystem. It is suggested that blockchain technology is a useful remedy for issues with scalability and security. For IoT-based systems, the suggested technique, CBcA, performs authentication and block transmission. Compared to PoW (proof of work), CBcA (confirmation of blockchain authentication) uses less resources for block validation and detects attacks faster. When using other techniques, the authentication procedure takes 10–13 s; however, with CBcA, it just takes 5 s at minimum. In comparison to other approaches, the suggested solution exhibits a respectable level of scalability. Blockchain technology is used by the proposed CBcA technique to encrypt communication between IoT gadgets. The suggested approach has quicker authentication procedures, less resource use, and more scalability [10].

Most of the IoT systems to control access now in use employ centralized designs and traditional identity management schemes. These methods and architectures have identified security and privacy concerns, such as the risk of single points of failure or monitoring, like device tracking [11].

The authors of the study [12] combined identity-based signature (IBS), attribute-based control of access (ABAC), and permissioned blockchain (HLF) to create a cross-domain, secure, and lightweight blockchain-based IoT system to control access. In particular, authors separated the Internet of Things system into various IoT domains or function domains. In order to allow additional IoT gadgets to function as blockchain nodes, the authors then created a local blockchain ledger for each IoT domain. The characteristics of the IoT domain entities, policy file digests, and access choices are all recorded in the local blockchain ledger.

The study [13] commenced with a synopsis of blockchain and IoT and looks into the problems associated with IoT blockchain applications. In addition, this paper reviewed the most relevant jobs to analyze how blockchain technology might enhance IoT security and looked closely at present study problems and advancements in the application of blockchain-related methods and tools.

The study [14] outlined the main difficulties IoT systems face and how blockchain technology is suggested as a potential solution. Additionally, it assessed the state of recent deployment phases and research being conducted on the integration of blockchain technology with IoT networks. It also covered the problems with the actual IoT-blockchain integration. Ultimately, this study offered an architecture plan that uses cloudlet computing and dew to link blockchain with IoT in two levels. The authors' goal was to take advantage of blockchain's features and services to ensure decentralized data processing and storage, solve issues with security and anonymity, and accomplish openness and effective authentication.

A safe and effective smart home architecture that combines blockchain and cloud computing skills for a combined solution was proposed in the paper [15]. Blockchain technology's decentralized structure allows it to conduct transactions and create transaction copies of the sensible user data that is gathered via smart homes. The suggested model in this paper used the multivariate correlation analysis technique to examine network traffic and determine the correlation between traffic features in order to guarantee the security of smart home networks. The performance of our suggested architecture was assessed by the authors using several metrics, such as throughput, and they found that blockchain provides an effective security solution for the network of the future Internet of Things.

When compared to centralized and blockchain-based authentication techniques, the suggested solution for IoT provides considerable gains in terms of computing cost, execution time, and power consumption, according to the results in [16]. The author's design was found to be capable of mitigating threats and meeting IoT security standards through a security study.

In order to forecast diseases, the research in [17] suggested a safe healthcare service that uses blockchain and fog computing. It focuses on diabetes and cardiovascular conditions. Health data about patients is composed commencing the fog nodes and safely lying on blockchain. After grouping patient health data using a unique rule-based clustering approach, FS-ANFIS is used to forecast diseases.

In [18], the proposed framework ensured a cohesive and secure system, addressing challenges faced by traditional systems. By combining ERP for unified data management, blockchain for security, IoT for real-time student data, context awareness for sensing and interpreting environmental changes, and cloud computing for scalability, the paper envisioned an efficient educational ecosystem.

In the study [19], the authors proposed a system that enhances data quality, data audit records, and transparency by utilizing blockchain and smart contracts in conjunction with key-aggregate searchable encryption (KASE) to meet security needs. Additionally, the authors suggested a technique that protects PHR data owners' rights to data subject consent when assigning multiple rights through attribute tokens.

In [20], the authors have proposed the ticket-based control of access method. Using ticket-based AC to remove the single point of failure in the tag ID-based pub/sub model, the authors provided a way to apply blockchain (BC) technology to each node and disperse the processing done by the TAAI. Reducing the load on each node can be achieved by having the blockchain handle authentication.

In the paper [21], the authors proposed Secure Smart Home Communications based on the Ethereum blockchain and smart contract as a new design to solve the dilemma between the limited resources of IoT gadgets and the concerns of a centralized architecture. Quantitative and qualitative evaluations of the architecture under common threat models have highlighted its effectiveness in providing security and privacy for IoT applications.

Blockchain technology, which is characterized by decentralization, transparency, and immutability, was introduced in the paper [22]. It can be applied to address issues with data security and confidentiality that IoT gadgets face. The study on the role of access control (AC) techniques and blockchain technology in Internet of Things systems was also included in the paper.

A patient-centered technical architecture for managing chronic illnesses with wearable, blockchain, and artificial intelligence was presented in the paper [23]. The building that was suggested was based on the Metaverse setting. Patients and doctors must register on the blockchain network in order to engage in the Metaverse.

The proposed study in [24] investigated the problems and obstacles and how blockchain technology offers security and transparency in Internet of Things applications.

Blockchain (BC) and Information for Operational and Tactical Analysis (IOTA) are distributed ledgers that use decentralized databases to record a vast number of transactions simultaneously in many locations. In order to address the fundamental issues with IoT, edge computing (EC) was developed. It offers real-time processing, resource management, and storage services closer to IoT gadgets on the edge of the network. The survey in [25] described the drawbacks of IOTA and blockchain in EC and offered avenues for future research.

In the paper [26], the cloud computing concept, often known as the BCOT paradigm, is examined through a range of real-world applications. It covered the topic of blockchain's involvement in supply chain management applications. The main obstacles to blockchain integration with cloud and IoT are listed in depth.

The study in [27] presented DAC4SH, a decentralized and trustworthy smart contract-based control of access system for s-homes. To implement fine-grained data control of access, the suggested system included contracts for access policy management (APMC), data attribute management (DAMC), subject attribute management (SAMC), and data access control (DACC). The authors determined that the performance of DAC4SH is appropriately rooted in the results of the trial.

In the paper [28], the authors proposed a consortium blockchain in place of the conventional key authority, which solved the key escrow issue with a revocable blockchain-aided ABE with escrow-free (blockchain-ABE-EF) system.

The proposed approach in [29] has been demonstrated to enable accurate real-time analysis of vulnerable behavior. Based on the experimental simulations, the suggested solution effectively facilitated the development of superior logistic monitoring.

The experimental results in [30] suggested that the proof of concept that was produced shows that it can manage and protect IoT data in real-time effectively, maintain optimal transaction throughput and latency, and handle a variety of loads. Additionally, it guarantees ongoing synchronization with the blockchain network and data collection from IoT nodes in the supply chain.

In the paper [31], the authors presented a spectrum sensing method for the uplink communication scenario in NOMA with multiuser interference, which aims to make effective use of both static and dynamic gain on spectrum efficiency. Firstly, the sensing criterion and workflow are outlined. The closed-form solution for the sensing threshold configuration is derived thereafter.

In [32], the system improved efficiency, decreased energy consumption, and lessen spoilage risk by learning and evolving its techniques using RL applied to historical and real-time data. The IoT RL method improves cold chain logistics performance via a simulation-based assessment. Improvements in the control of temperature and humidity for perishable items have decreased spoilage and improved the dependability of the supply chain. It presented some challenges in cold chain logistics, such as confidentiality of data, dependability of sensors, and processing complexity of algorithms.

The main objective of the research work in [33] is to show how blockchain-based systems for billing in smart grids can offer a more advanced foundation for energy management. They are very efficient, secure, and open unlike any other option within that industry. Therefore, blockchain technology and smart contracts can be used by major stakeholders in the energy industry to improve their billings and streamline operations that they do. Henceforth, this sets grounds for a sustainable and elastic future of energy.

The study in [34] presented an innovative method for improving privacy protection in medical gadgets based on the Internet of Things (loT). It achieved this by combining homomorphic techniques for encryption with blockchain technology.

By integrating IoT and machine learning technologies, the innovative waste management system in [35] enhanced operational efficiency, promoted sustainability through reduced waste accumulation, and improved overall cleanliness in hostel mess environments.

The paper in [36] implemented delegated verification service using Intel Software Guard Extensions (SGX) as TEE. The evaluations of the implementation showed that the author's proposal is scalable and practical with excellent runtime performance.

In [37], authors proposed a secure data-sharing framework that uses local differential privacy (LDP) within a permissioned blockchain, enhanced by federated learning (FL) in a zero-trust environment. To further protect sensitive data shared by IoT gadgets, the authors used the IPFS and cryptographic hash functions to create unique digital fingerprints for files.

In [38], the project team conducted research and analysis on control of access between geomagnetic observation gadgets based on fog computing combined with blockchain technology. The IoT control of access model suitable for geomagnetic observation network systems was designed based on research results.

The paper in [39] presented a new hybrid modular framework incorporating IoT technologies and blockchain. One of the proposed framework's main goals is to eliminate the flaws related to drone communication networks, including security risks and performance inefficiency.

From the review findings, the study in [40] gave possible suggestions for the future trend of IoT. With this review, both the manufacturer/production industry will be able to affect the interactions and the progress of global connection, and the Internet of Things will be greatly aided by major technologies, including machine learning (ML), artificial intelligence (AI), 5G blockchain technology, and cloud computing at the edge.

The study in [41] pays particular attention to the emergence of the industrial revolution and the current trend in the development of IoT-enabled medical gadgets for various healthcare systems use. It was discovered that IoT technology has several advantages for the creation of medical gadgets, including better data sharing and communication, higher precision and safety, and lower costs for the deployment and upkeep of gadgets.

The proposed method in [42] makes use of a ResNeSt model with Split-Attention (ResNeSt), which is enhanced by a GRU and RSG-MJ. The results of the testing indicate that this technology is a good fit for real-world IoT security problems since it can detect DDoS attacks more quickly, which improves performance.

The authors of the work proposed in [43] the use of blockchain technology to ensure data safety, as well as the option of building ITS architecture with a module for monitoring incident data and timely response. This paper provided mathematical calculations of the opportunity of using blockchain technology within the framework of the proposed concept and modeling.

In [44], the authors presented the modeling and implementation of two urban use cases for a six-stage IoTinuum: smart drone delivery and smart structural monitoring. In the author's experience, the IoTinuum improves the understanding of function placement in application development and reveals important performance trade-offs.

In paper [45], the authors proposed a novel architecture that combines sensor, drone, blockchain, and machine learning technologies to support a long-term monitoring system for vast areas of land. Long-term monitoring is achieved by employing multigenerational sensors that have been deployed at the start but are woken up in stages as and when needed to replace or reinforce existing sensors. A swarm of drones periodically scans the landscape to identify areas that need immediate attention.

The authors proposed in the paper [46] a distributed VPN-based solution that relies on distributed fog nodes playing the role of VPN servers. To determine the VPN server responsible for a specific communication, a fully decentralized and efficient search is guaranteed within the fog nodes through the utilization of the DHT-based Chord protocol. To enforce security and immutability, blockchain is used to verify the different criteria of a new requester to join a specific VPN.

The proposed solution in the paper [47] offers robust security and scalability in electronic data storage, providing users with control over their information through a blockchain-based system. The system presented a promising solution in preventing data breaches and safeguarding sensitive healthcare information.

The research in [48] introduced a proof of concept for a lightweight, scalable, and modular blockchain network using Hyperledger Fabric to enhance federated learning. Performance evaluations using Hyperledger Caliper showed that the proposed blockchain network effectively improves federated learning without compromising latency and throughput.

The implementation of the proposed algorithms in [49] validated the resilience of the proposed approach against cyberattacks. Overall, the proposed approach enables efficient and secure D2D communication for resilient SDN infrastructure in IoT ecosystems.

A method for protecting consumer electronics gadgets in Internet of Things ecosystems called blockchain-based access management with DL threat modelling (BCAM-DLTM) was described in the paper in [50]. Threat detection and access management are the two primary phases of the BCAM-DLTM technique's operation. Additionally, consumer electronics device access management can be implemented using blockchain technology. Furthermore, the BCAM-DLTM method uses a deep belief networks (DBNs) model to effectively detect threats. The reptile search algorithm (RSA) is used in the hyperparameter tuning process to improve the DBN model's recognition performance. The NSLKDD dataset is used in the BCAM-DLTM approach's experimental outcome research.

The secure computation offloading system model in IoT-fog-cloud systems with blockchain is proposed, and a secure computation offloading model is proposed. It adopts deep reinforcement learning (DRL) and smart contract for latency, energy, and cost efficiency. Supporting on-demand operation, online and offline modes, this system gains efficiency gain and outperforms four baselines with respect to energy consumption, latency and task failure rate [51].

A blockchain-IoD-based RBFNN model is provided for the security of IoD. The model improves intrusion detection and decentralization of deep learning in drone networks. The evaluations demonstrate that XCIM can improve the data integrity and significantly outperforms other methods in the precision, recall, and F1 score, which are useful for intelligent and secure drone communication [52].

In this paper, the tolerance of failure in wireless sensor networks (WSNs) is investigated on the basis of fault trees and Markov chain analysis. By studying system behavior under different failure conditions, the book also provides you with practical guidance on how to improve the dependability and fault tolerance of the system, helping you to develop systems that are better suited to industrial WSN applications [53].

IoT is a recognition of itself through devices that are supported by a wireless system. This highlights the difficulty of integration, the shortcoming of monitoring standards, and issues that have to do with security. Approaches such as those defined with genetic algorithms and PSO are categorized using MATLAB and Python as commonly used software. The key parameter is the adaptability: it gives hints for future study [54].

Deep learning for climate change applications are as follows: A systematic review of bad, better, and best methodologies, a systematic literature review on deep learning for climate change mitigation, and a scoping review over three areas: ecosystem, agriculture, and industry. It presents a new taxonomy and evaluates models in terms of accuracy, scalability, and interpretability. Applications are overwhelmingly dominated by CNNs and Python. The results inform future sustainable solutions and identify weaknesses in terms of data integration and model implementation [55].

The paper investigates the use of deep learning for diagnosis of Alzheimer's disease. Methods incorporating CNNs and NLP are promising for the diagnosis and treatment. Meanwhile, the explainability and security of the data are also issues. Python is a popular language, and the authors highlight the importance of interpretable and secure AI to drive future developments in AD diagnosis [56].

This study classifies autonomous learning applied in inter-domain systems such as security and healthcare. They emphasize the absence of a community-wide taxonomy and emphasize issues of precision, privacy, and scalability. It is typical to use Python and MATLAB. This work can be viewed as a showcase for more systematic research on adaptive, domain-specific intelligent systems [57].

In the fourth wave of human evolution (cyber-age), the prediction is a pan-global connectivity between all things via the IoT. It enables people and things to communicate globally in real time with minimal human intervention and is transforming how society operates and interacts by embedding a digital fabric throughout its computing infrastructure [58].

In this paper, we suggest a QCA-based ALU (arithmetic logic unit) for secure IoT systems. The ALU architecture achieves speed enhancement and power reduction and is less area-consuming. The resulting simulations show a 60% reduction in area and cell count with respect to similar models from the literature, and it proves to be a high performance and secure alternative to usual hardware-based implementations [59].

This paper introduces a blockchain-based access control system with smart contracts to secure sharing of EHRs. It guarantees data security on mobile healthcare, including data streaming from sensors. A high-security, low-latency, and high-stealth network framework is essential for trustworthy health data transmission [60].

This paper is to suggest a blockchain-secured 6G wireless network through which the DoS attacks can be prevented by the use of reinforcement learning and optimization algorithms. It has achieved excellent results in throughput, power consumption, and inaccuracy. The simulation results verify that the new approach is suitable for secure and efficient 6G communications with path selection and shorter network delays [61].

An IoT and adaptive autoencoders aided hybrid deep learning model is introduced for 5G mmWave communication. It enhances network slicing and routing using deep clustering methods. The performance of the model achieves over 97% accuracy as tested by cross-validation and performs well on important indicators as packet delivery, latency, and precision [62].

The research constructs a NLP/statistical ML-based cyberbully detection system. It uses Twitter and Wikipedia data sets, and it estimates classifiers: SVM or random forest. These models were able to accurately identify hate speech and offensive comments with approximately 90% accuracy, leading to more reliable content moderation on social media [63].

To address the security of healthcare data transmission in IoT systems, we present a blockchain-based solution. Trust-weighted PoA consensus, DBSCAN classification, and advanced encryption are the methods used, and the model can obtain an accuracy of 99.38% with low latency. It tackles limitations of classical approaches and enhances the scalability, transparency, and credibility of data exchange [64].

In this paper we present a new Glowworm Swarm Optimization (GSO)-based strategy to improve the performance of blockchain architecture for supporting IIoT. The approach enhances the allocation of resources, efficiency, and the scalability. Simulations also demonstrate considerable improvement over classical models, which makes it feasible for secure high-performance industrial IoT [65].

The survey summarizes botnets lifecycle, detection, and evasion strategies. It examines their use of covert channels and encrypted communication to escape detection. The paper enumerates the challenges in the fight against botnets and presents research directions that could be used in the future to make security control architectures more effective against such dynamic threats [66].

Regarding the fusion of industrial IoT and blockchain, the paper proposes a Hyperledger Sawtooth-based framework. It provides the secure node-to-node transactions and power-efficient communications. Leveraging pseudo-chain codes and consensus mechanism, our solution provides trusted data exchange in a resource-limited environment and is proven to be practical through industrial deploy simulation using Docker [67].

From the above study, the authors have concluded that different authors proposed different IoT access control systems using blockchain and tried to advance the safety of the structure. Based on the literature survey here, we have described some consecutive architectures that have evolved yet.

In Fig. 1, we have an IoT network of 100 + gadgets. There are a number of databases that store data generated

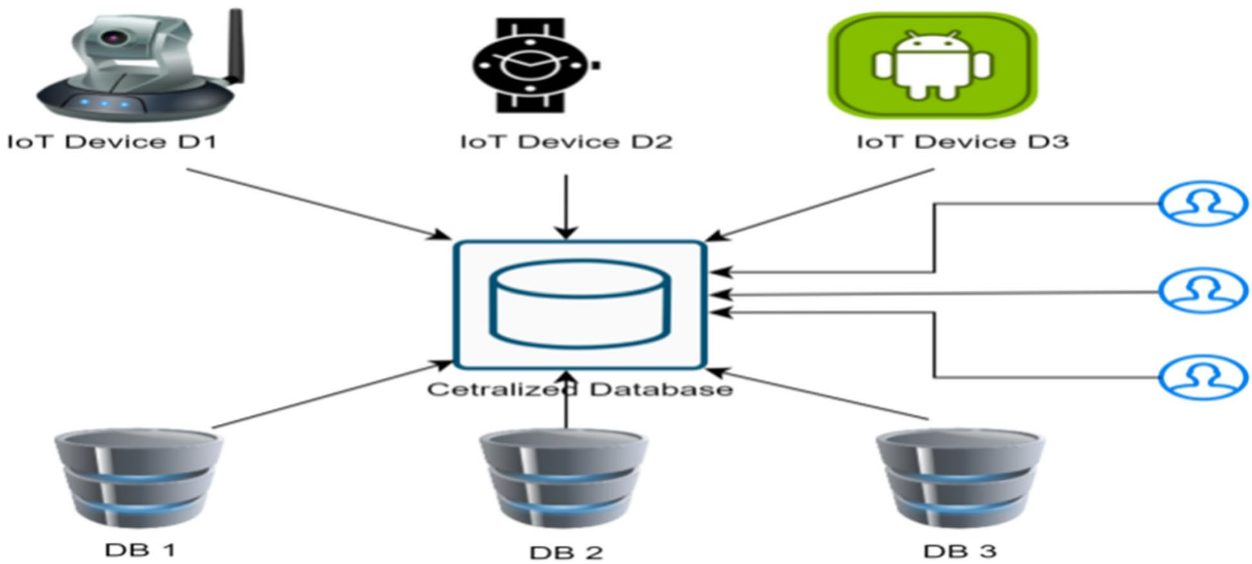

Fig. 1 IoT network

by the IoT gadgets, and that data can be accessed by the users according to their requirements.

Different types of IoT gadgets generate different kinds of data, like motion detection, temperature, humidity, and face detection. A separate database must store this data. A centralized database system connects all these gadgets and databases. Any user can access any type of data through the centralized database. As the number of users increases, the load on CDB will also increase. The main disadvantage of this type of network is the single point of failure. If CDB is down or having some serious issues, then it will affect the whole network. To solve this problem, we can put multiple copies of CDB so that if one server is facing some issue, a request can be sent to another server. However, if a hacker manages to compromise the CBS and steals confidential files from CDB, it could potentially harm the entire network. So, to solve the security issue of an IoT system, we want to integrate the blockchain with an IoT system, as shown in Fig. 2. Due to the unique features of private blockchain, it will provide a strong, secure, decentralized, and scalable IoT system; without permission, no one can enter the network.

The integration of blockchain with IoT still faces significant scalability challenges, such as limited transaction throughput and high latency, which hinder its use in large-scale environments. While research has focused on blockchain's security in IoT, solutions for scalability remain underexplored. Addressing this gap requires developing lightweight, scalable blockchain architectures, possibly through hybrid consensus mechanisms or off-chain scaling techniques, to enable efficient IoT applications. The device manager, user manager, and database manager are not decentralized; they operate at the application layer and are connected to the blockchain. These components handle device authentication, user access management, and structured data storage, ensuring smooth interaction with the blockchain for secure data transactions. While blockchain ensures data integrity and access control, these managers function as centralized entities that facilitate efficient data processing and system operations.

2.1 Research gap

Despite extensive work, existing architectures either suffer from single points of failure or impose significant overheads on IoT devices. While blockchain improves decentralization and transparency, its throughput limitations remain a bottleneck. Few studies explore the integration of asynchronous messaging layers like RabbitMQ with blockchain for access control this paper addresses that gap by introducing a hybrid architecture that leverages ABAC policies, smart contract.

2.2 Problem statement

In this paper, we focus on identifying the security and scalability of an IoT system by studying approximately 50 research papers and investigating that an IoT system is dependent on a centralized server and it has some issues like single point of failure, throughput, and fewer resources. And to overcome the problems of an IoT system, we are using private blockchain, and to improve the scalability of an IoT system, we are using message queuing systems (Table 1).

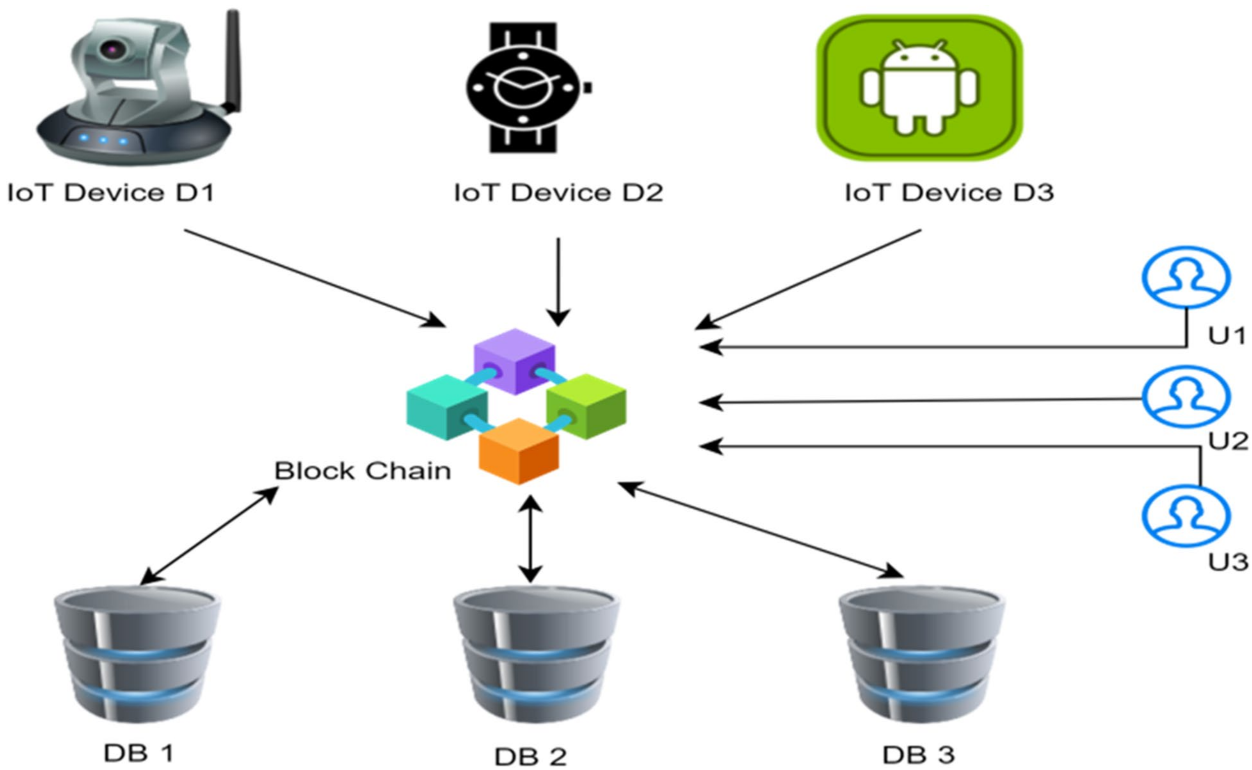

Fig. 2 Decentralized IoT network

Our study considers the generic IoT scenario with a wide influence on various domains like Industrial IoT (IIoT), Internet of Vehicles (IoV), smart cities, and healthcare IoT. Our blockchain design is built as domain-independent system which can support data access protection for various IoT implementations with the same level of security guarantees. The modular and flexible architecture can address the specific security requirements and constraints faced by various IoT domains. This holistic model will enhance the wider impact and relevancy of the research contribution within the highly fragmented and fast-growing IoT ecosystem.

3 Proposed model

3.1 Enhancing IoT systems scalability using blockchain and RabbitMQ

3.1.1 Overview

Blockchain technology, while inherently secure and decentralized, faces significant challenges when it comes to scalability. As transaction volumes increase, the system can experience latency and congestion, limiting its potential for large-scale applications. To address this, we propose a robust message-broker system to optimize transaction processing and block propagation across blockchain nodes, ultimately enhancing overall scalability.

The message queuing system is necessary in the management of high volume, asynchronous transactions on the IoT, where stemming decision requests can then be issued to potentially thousands of devices at once which would not be able to be handled with synchronous blockchain-only method. Fault-tolerant message handling, load balancing among the set of access control nodes, and separation of the blockchain processing from the real-time requests to the IoT devices enable the responsiveness also at the times of heavy traffics load. In addition, other benefits of the asynchronous queuing model are achieved in our framework, such as maintaining service availability under temporary blockchain network delays or node failures, which becomes critical for continuous IoT operations and is validated by our experiments with better execution time and system scalability.

3.1.2 RabbitMQ as a scalable messaging layer

RabbitMQ operates as an intermediary between nodes in the blockchain network, facilitating the efficient exchange of transaction data and blocks. By decoupling communication between nodes and using RabbitMQ's asynchronous messaging system, we can significantly reduce the network overhead and ensure a more reliable and scalable message flow. RabbitMQ's features, such as queue management, load balancing, and fault tolerance,

Table 1 Comparative study of different architecture

	[4]	[5]	[6]	[7]	[12]	[8]	[9]	[2]	[3]	[10]	[13]	[14]	[15]	[1]	**Proposed arch**
Type of arch	DC-ID	DC		DC										DC	BC + MQ + ABAC
Technology used	HLF + BC	BC	BC + storage + cryptography	Smart contract	HLF + IBS + ABAC	Local peer + CA	Fog/edge server + SC + sidechain	Onchain, offchain	Cluster	CBCA	BC	BC + IoT + dew layer + cloudlet	BC + SH	BC	BC + MQ
Access control	ABAC			TBA					MCCAC						TBAC
Authentication				Yes			Yes			Yes					Yes
Efficiency		Yes												Yes	
Security		Yes	Yes		Yes						Yes		Yes	Yes	Yes
Feasible														Yes	
Scalable	Large scale					Yes	Yes			Yes					Yes
Reliability			Yes												
Lightweight					Yes										
Throughput	High						Yes	Yes							
Communication cost			Yes				Yes								
DDoS attack					Yes										

are essential to improving the efficiency of blockchain systems.

3.1.3 Architecture and design

The proposed architecture introduces RabbitMQ as a middle layer between blockchain nodes as follows:

- *Message queues*: Transactions and block data are propagated through RabbitMQ queues. Each node subscribes to the relevant queues and processes transactions or blocks in an asynchronous manner. This approach reduces the need for direct peer-to-peer communication, minimizing the bandwidth required for block propagation.
- *Exchanges*: RabbitMQ's exchange system is used to route transactions and blocks to the appropriate queues based on node availability and type. We use different types of exchanges (direct, topic, or fanout) depending on the blockchain's consensus mechanism and the type of data being propagated.
- *Node scalability*: New nodes can be added to the blockchain network without disrupting the existing system. RabbitMQ dynamically adjusts to accommodate new queues and exchanges, ensuring balanced message delivery even with fluctuating transaction volumes.

By leveraging RabbitMQ, the blockchain system benefits from the following:

- *Asynchronous processing*: Transactions and blocks are processed asynchronously, reducing the wait times and allowing nodes to perform tasks concurrently, improving throughput.
- *Load balancing*: RabbitMQ's load-balancing capabilities distribute the workload evenly across nodes, preventing any single node from being overwhelmed with excessive transactions or block validation tasks.
- *Dynamic scalability*: As transaction volumes increase, RabbitMQ scales by dynamically adding more queues and exchanges to handle the load, ensuring that the system can accommodate higher transaction volumes without performance degradation.
- *Efficient communication*: By replacing direct peer-to-peer communication with RabbitMQ's message-routing system, we reduce the communication overhead typically associated with blockchain systems.

The algorithm outlines a scalable IoT system using blockchain and RabbitMQ for efficient transaction management. First, it initializes the blockchain network and RabbitMQ as a message broker to handle communication. Incoming transactions are distributed to different RabbitMQ queues based on type or workload, and blockchain nodes pull transactions from assigned queues for processing. Parallel transaction processing is achieved by optimizing queue assignments. Blocks are propagated via RabbitMQ exchanges once validated. The system includes dynamic load balancing to handle queue overloads and allows horizontal scaling by adding more nodes and queues based on network load.

Algorithm: IoT System Scaling with Blockchain and RabbitMQ

```
Input:
- Transaction Set: `T = {t1, t2, t3, ... tn}` (Incoming blockchain transactions)
- Blockchain Nodes: `N = {n1, n2, n3, ... nm}` (Distributed blockchain validators)
- Message Broker: `R` (RabbitMQ for task distribution and communication)

Output:
A highly scalable, low-latency blockchain system leveraging RabbitMQ for intelligent workload distribution, parallelized transaction processing, and real-time load balancing, ensuring efficient handling of high transaction volumes in decentralized environments.

Step 1: System Initialization
1. Deploy blockchain network with nodes `N`.
2. Configure RabbitMQ as the message broker `R` for decentralized communication.
3. Establish dynamic routing rules for efficient message queuing.
Step 2: Intelligent Transaction Distribution via RabbitMQ
1. Capture real-time transactions `T` from clients.
2. For each transaction `ti ∈ T`:
  - Categorize transactions based on type and priority.
  - Publish `ti` to RabbitMQ queue `Qi`, ensuring adaptive load balancing.
Step 3: Blockchain Node Queue Subscription & Processing
1. Each node `ni ∈ N` subscribes to designated RabbitMQ queue(s).
2. Nodes pull transactions `ti` from `Qi` in a distributed manner.
3. Execute validation and processing steps:
  - Implement lightweight consensus mechanisms (e.g., PoS, PBFT) for faster transaction finality.
  - Construct block `Bi` once a sufficient number of transactions are validated.
Step 4: Parallelized and Adaptive Transaction Processing
1. Enable concurrent processing across multiple nodes for scalability.
2. Optimize transaction queue assignment via:
  - Adaptive round-robin scheduling to evenly distribute load.
  - Priority-based queuing for time-sensitive transactions.
  - Dynamic sharding to minimize processing latency.
Step 5: Secure and Efficient Block Propagation
1. Once a node `ni` finalizes block `Bi`:
  - Broadcast `Bi` to the network using RabbitMQ exchange `E`.
2. Nodes validate received block `Bi`:
  - Execute rapid consensus verification.
  - If valid, append `Bi` to the local blockchain.
Step 6: Autonomous Load Balancing and Network Scaling
1. Monitor real-time RabbitMQ queue metrics.
2. If queue `Qi` shows congestion:
  - Dynamically reallocate computational resources across nodes.
  - Auto-scale network by provisioning additional blockchain nodes.
  - Implement intelligent workload distribution based on historical transaction trends.
Step 7: Continuous Performance Optimization & Scalability Assessment
1. Periodically analyze RabbitMQ queue throughput.
2. Adjust processing power or queue allocation dynamically to optimize throughput.
3. Scale horizontally by:
  - Deploying additional RabbitMQ queues for multi-tier processing.
  - Expanding blockchain nodes to sustain increasing transaction loads.
```

The algorithm integrates attribute-based access control (ABAC) with blockchain to manage secure, immutable access control. First, attributes like user role and clearance are defined, and smart contracts containing ABAC rules are deployed on the blockchain. When a user requests access to a resource, their attributes are authenticated and logged as a blockchain transaction. Smart contracts evaluate the user's attributes against the policies to approve or deny access. The access decision is logged immutably on the blockchain for future auditing, ensuring transparency and accountability in access control decisions.

Algorithm: Attribute-Based Access Control (ABAC) with Blockchain

Input:

-User Attributes: `{role, department, clearance_level, etc.}`
- Resource Request: `{resource_id, requested_action}`
- Blockchain Smart Contract:`{Access Control Policies}`
- Identity Provider:`{Trusted Source for User Attributes}`

Output:
A secure, decentralized, and auditable access control system using blockchain-based ABAC, ensuring tamper-proof access policies, real-time authorization, and transparent compliance tracking.

Step 1: System Initialization
1. Define Access Attributes:
 - Identify key attributes governing access control (e.g., `role`, `department`, `clearance_level`).
2. Deploy Smart Contract:
 - Implement and deploy ABAC-based smart contracts on the blockchain.
 - Encode rules defining required attributes for each resource access.
3. Set Up Blockchain Network:
 - Configure the blockchain infrastructure to store access logs and policies immutably.
Step 2: User Request Handling
1. Authenticate User:
 - Retrieve user attributes from a trusted identity provider.
2. Submit Access Request:
 - User submits request `{resource_id, requested_action}` with associated attributes.
3. Log Request as Blockchain Transaction:
 - Store the access request as an immutable transaction on the blockchain.
Step 3: Access Control Enforcement
1. Evaluate Smart Contract:
 - The smart contract retrieves the user's attributes and compares them against stored policies.
2. Policy Verification:
 - If user attributes match the policy conditions, access is granted.
 - Otherwise, the request is denied.
Step 4: Access Decision & Logging
1. Grant or Deny Access:
 - If approved, the user gains access to the requested resource.
 - If denied, the user receives a rejection message.
2. Record Access Decision:
 - Log the access decision on the blockchain for future auditing.
Step 5: Auditing and Accountability
1. Immutable Storage:
 - Store all access requests, policy checks, and decisions immutably on the blockchain.
2. Audit Trail for Compliance:
 - Allow authorized auditors to trace and verify access control decisions.
 - Ensure transparency and compliance with security policies.

Blockchain is used; it is a secure, decentralized, and immutable ledger, which secures data more firmly and provides proper access of data to the authorized user. But before moving further, we should notice two important points to access any system, i.e., authentication and authorization.

Decentralized IoT system architecture in Fig. 3 includes three components: IoT gadget, users, and database. IoT gadgets generate data like temperature, motion detection, and humidity; these data must be saved in respective valid databases. To properly authenticate users and gadgets, there is one user manager who will take care of user registration, authentication, and authorization. The device manager ensures the device registration and authentication, respectively. All these three components are connected to each other through a blockchain network. Access rules are written in the form of a smart contract, which is stored in blockchain. A self-executing program that automates the acts specified in a contract or agreement is known as a smart contract.

Figure 3 shows that the user managers handle all the user-related activity, like user registration, user authentication, and authorization. Similarly, device managers handle all device-related activities like device registration and device authentication. Likewise, the database manager will manage access to the database, including reading and writing to the database. In such a situation, if the number of requests to access DB increases to some extent, there may be a chance that the request may have to wait for a long time, or sometimes request may be lost due to bulk message.

So, to avoid the situation in the proposed architecture, a message queuing system (Fig. 4) is used. In a message queue, requests might have to wait for a little while, but they will never get lost. It is the unique feature of the message queue system that each and every message is properly delivered and acknowledged. Until and unless we receive the acknowledgement for the previous message, the next message will remain in the queue. As soon as we receive acknowledgement, then the next message will be processed. For example, in the [11] e-voting system, there are two voters named Shaun and Ferry who want to give their votes. Shaun's voting system is connected to a web server, and Ferry's voting system is connected to a message queuing

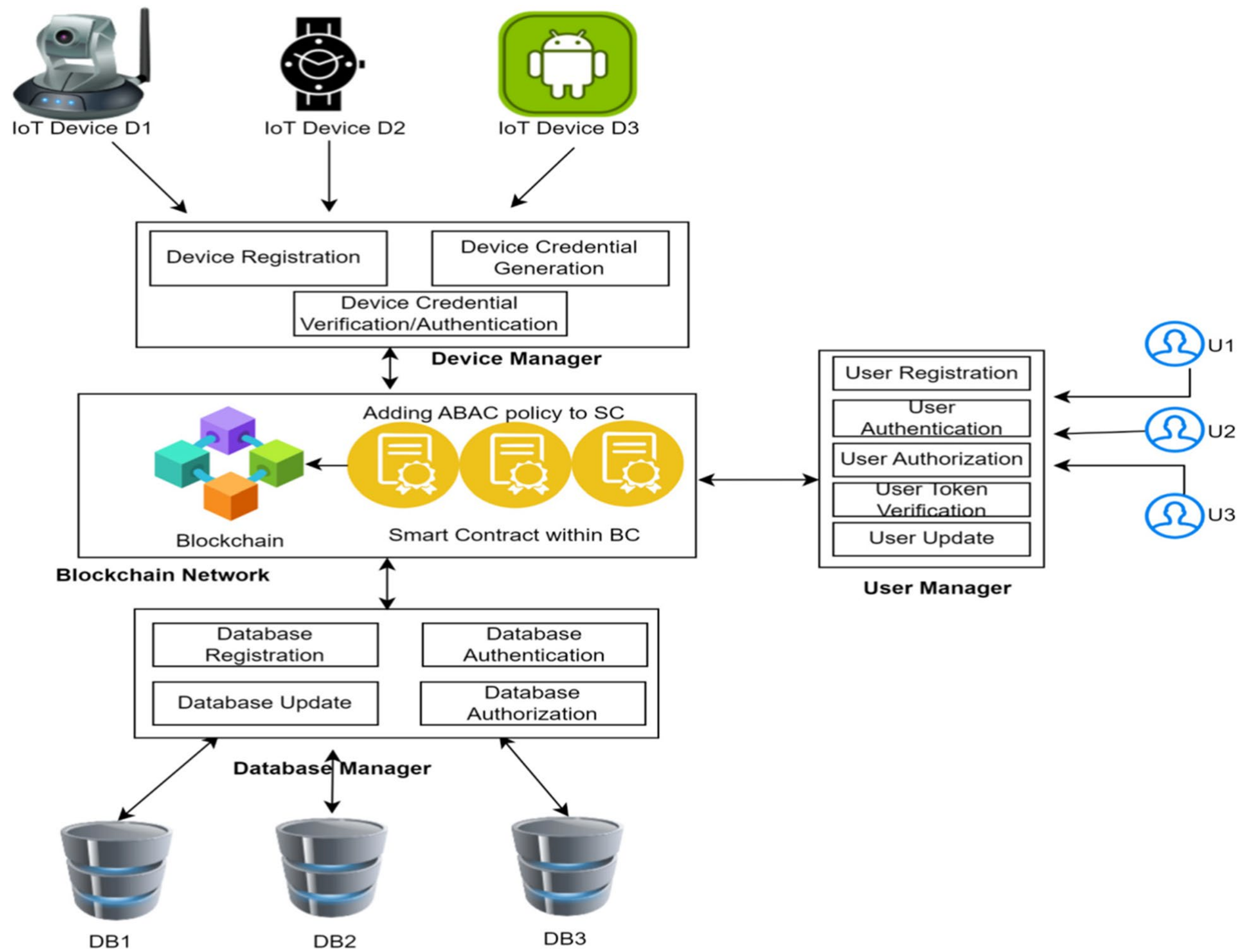

Fig. 3 An IoT access control system using blockchain

system, so when they gave their vote, they received the acknowledgement on their phones. But if there is some network issue or server unavailability, then in the case of Shaun, he will receive the message that the server is not available. After some time, he can retry for vote submission, and he will deliberately obtain the privilege to change his voting choice. But in the case of Ferry, messages will remain in the queue on the client side in case of a network issue, and when the server is back, the message is pulled out of the queue and processed. And after successful submission, Ferry receives acknowledgement on her phone. She does not need to retry her submission, so this is the beauty of the message queuing system: messages never get lost.

3.2 Pseudocode added

3.2.1 Pseudocode for using RabbitMQ to handle IoT operations with blockchain integration

The code now includes validating requests through blockchain smart contracts for saving and fetching data and user authorization.

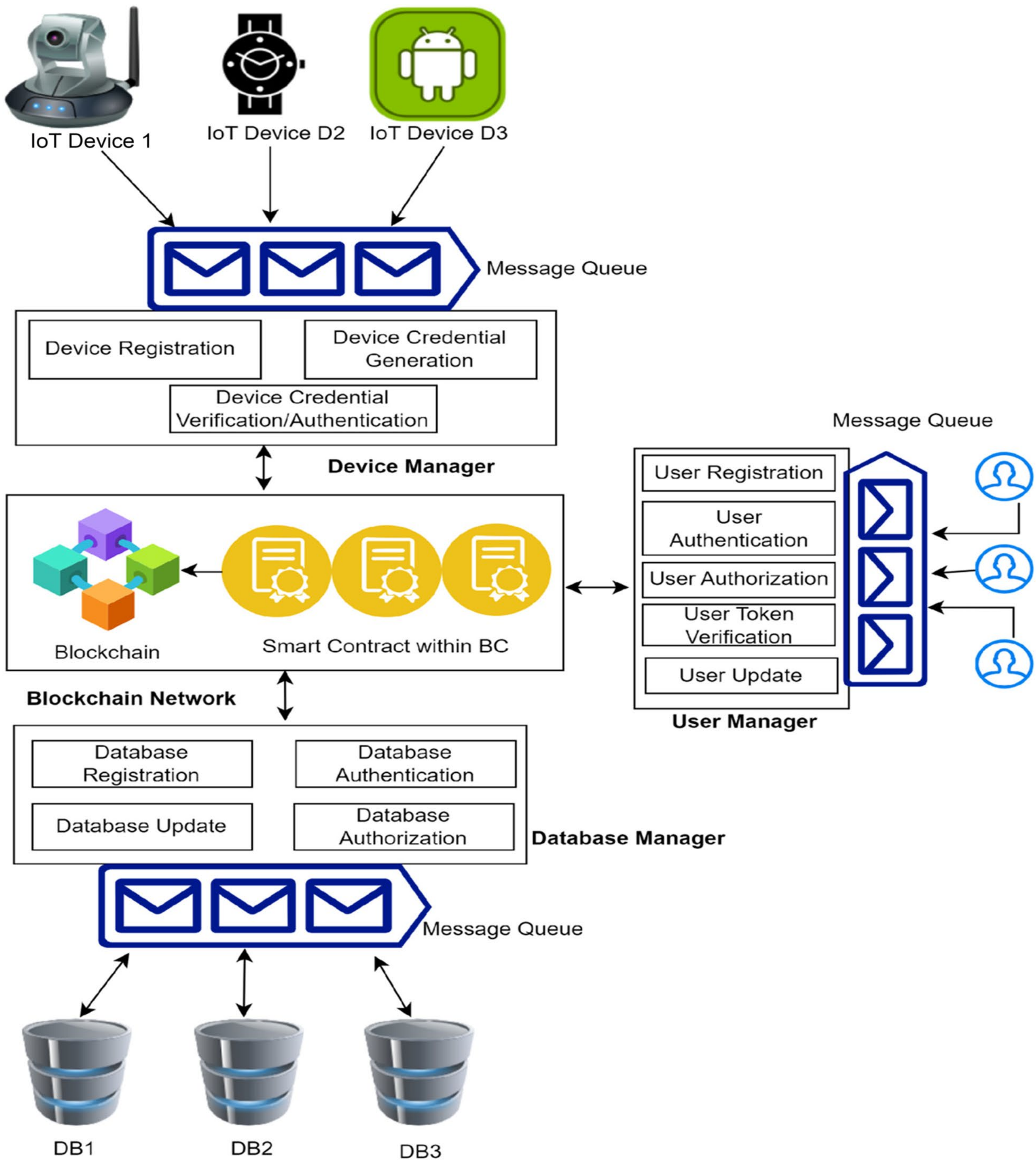

Fig. 4 An IoT access control system using blockchain and message queuing system

Algorithm: Scalable Blockchain-Based IoT Message Processing Framework

```
Step 1: System Initialization and Parameter Definition
1. Define IoT Network Parameters:
   - Set `N      100` (Total IoT devices).
   - Define message production rate per device: `λ_i      1/5` messages per second.
   - Compute total message production rate: `λ_total      N * λ_i`.
2. Define Consumer Processing Constraints:
   - Set `C      5` (Total message consumers).
   - Define message processing rate per consumer: `μ_c      5` messages per second.
   - Compute total message consumption rate: `μ_total      C * μ_c`.
3. Set Failure Probability:
   - Define probability of message processing failure: `p_f      0.05`.
Step 2: Dynamic Message Generation for Blockchain IoT Transactions
1. Simulate Message Production:
   - For each `device_id` from `1` to `N`:
     - For each time step `t`:
       a. If `(t % 5 == 0)`, create a new message.
       b. Publish message to `'blockchain_exchange'` with routing key `'save_data'`.
Step 3: Intelligent Exchange Routing for Blockchain Transactions
1. Function `publish_message(exchange, routing_key, message)`:
   - If `exchange == 'blockchain_exchange'`:
     . If `routing_key == 'save_data'`, forward to `'save_data_queue'`.
     b. Else if `routing_key == 'fetch_data'`, forward to `'fetch_data_queue'`.
Step 4: Adaptive Queuing and Efficient Message Processing
1. Initialize Message Queue:
   - Define `blockchain_queue`.
   - Compute queue utilization: `ρ      λ_total / μ_total`.
   - Initialize queue state variables:
     a. `L      0` (Expected messages in queue).
     b. `W      0` (Expected waiting time per message).
2. Function `route_message_to_queue(queue, message)`:
   - Enqueue message into `queue`.
   - Recompute `L` and `W` dynamically based on queue load.
   - Distribute messages among `C` consumers for parallel processing.
3. Function `consume_message(queue)`:
   - While `queue` is not empty:
     a. Dequeue a message.
     b. Process message asynchronously.
     c. If processing fails, invoke `handle_redelivery(queue, message)`.
Step 5: Intelligent Fault Handling and Redelivery Strategy
1. Function `process_message(message)`:
   - If `validate_request(message)`:
     a. If `message.type == 'save_data'`:
        - Store data securely.
        - Acknowledge successful processing.
        - Return `True`.
     b. Else if `message.type == 'fetch_data'`:
        - Verify user access.
        - Retrieve and transmit requested data.
        - Acknowledge message reception.
        - Return `True`.
   - Return `False` if validation fails.
2. Function `handle_redelivery(queue, message)`:
   - Reinsert message into `queue` for retry.
   - Apply exponential backoff mechanism to prevent congestion.
Step 6: Secure Blockchain-Integrated Data Management
1. Function `validate_request(message)`:
   - Authenticate message using a blockchain smart contract.
2. Function `validate_user_access(user_id, data_id)`:
   - Confirm user access rights via blockchain validation.
3. Function `store_data(data)`:
   - Commit structured data to the blockchain ledger.
4. Function `fetch_data(data_id)`:
   - Retrieve requested data from blockchain storage.
5. Function `send_response_to_user(user_id, data)`:
   - Deliver verified data securely to the requesting user.
```

This pseudocode leverages RabbitMQ for routing IoT data requests and blockchain smart contracts for validation and authorization. It ensures secure data management by validating storage requests and user access before performing operations. The system handles queuing, processing, and failures with error handling, integrating blockchain to maintain data integrity and manage authorization efficiently.

3.2.2 Pseudocode for implementing policy management in a blockchain-based smart contract system

Algorithm: Policy Smart Contract Implementation

```
 Step 1: Function __init__(api_stub):
        # Initialize API stub for blockchain interaction
        Set self.api_stub = api_stub

 Step 2:  Function validate_AS(as_section):
        # Validate AS section (Access Subject)
        Set required_attrs_AS = {"user_id", "role", "group"}
              If as_section is None OR not all required_attrs_AS in as_section:
  Return False
              Return True

 Step 3:  Function validate_AO(ao_section):
        # Validate AO section (Access Object)
        Set required_attrs_AO = {"device_id", "mac_address"}
              If ao_section is None OR not all required_attrs_AO in ao_section:
        Return False
Return True

 Step 4:  Function validate_AE(ap_section, ae_section):
        # Validate AE (Access Execution) and AP (Access Permissions)
        Set valid_values = {"1", "0"}
              If ae_section is None OR ae_section['value'] not in valid_values:
        Return False
              If ap_section is None OR ap_section not in valid_values:
        Return False
Return True

 Step 5 :  Function check_policy(policy):
    # Comprehensive Policy Validation
    If 'AS' not in policy OR NOT validate_AS(policy['AS']):
Log("Invalid AS section")
Return False
    If 'AO' not in policy OR NOT validate_AO(policy['AO']):
Log("Invalid AO section")
Return False
      If 'AE' not in policy OR 'AP' not in policy OR NOT validate_AE(policy['AP'], policy['AE']):
        Log("Invalid AE or AP section")
        Return False
Return True

 Step 6:   Function sha256(data):
        # Generate SHA256 Hash for Unique Identification
        Import hashlib
        Set hash_data = hashlib.sha256(data.encode('utf-8')).hexdigest()
        Return hash_data

 Step 7:   Function add_policy(policy):
        # Step 1: Validate Policy
        If NOT check_policy(policy):
        Return "Error: Invalid Policy Format"
              # Step 2: Generate Unique Policy ID
        Set policy_id = sha256(policy['AS'] + "+" + policy['AO'])
              # Step 3: Check for Duplicate Policy
        If self.api_stub.get_state(policy_id) is NOT None:
        Return "Error: Policy Already Exists"
             # Step 4: Store Policy in Blockchain
        Try:
        Import json
        Set policy_data = json.dumps(policy)
        self.api_stub.put_state(policy_id, policy_data)
        Log("Policy stored successfully with ID: " + policy_id)
        Catch Exception as error:
        Log("Storage Error: " + error)
        Return "Error: " + error
        Return "Success"

 Step 8 : Function delete_policy(policy_id):
        # Step 1: Check if Policy Exists
        If self.api_stub.get_state(policy_id) is None:
        Return "Error: Policy Not Found"
        # Step 2: Delete Policy from Blockchain
        Try:
        self.api_stub.del_state(policy_id)
        Log("Policy deleted successfully with ID: " + policy_id)
        Catch Exception as error:
        Log("Deletion Error: " + error)
        Return "Error: " + error
        Return "Success"
```

The pseudocode delineates a PolicySmartContract class that incorporates functions for the validation of policies, hashing procedures, and the management of errors. It verifies policy structure, generates unique IDs using SHA256, and manages policies with put_state and del_state. Error handling is included for robustness. This approach outlines a structured way to implement policy management on a blockchain.

3.3 Working of proposed architecture

1. If a user wants to access the data from DB, the user must be registered.
2. After user registration, authentication of the user is verified against information stored in blockchain.
3. Then the user is provided with the token; the user manager will verify the token, and if it is a valid token, then the user is authenticated.
4. The user manager also ensures which user can access which data or what access rights users have.
5. Access rules are defined in a smart contract, which is stored in the blockchain.
6. User authorization is verified by the user manager through information stored in blockchain.
7. After authentication and authorization of the user and devices, the user is allowed to access data (Fig. 5).

3.4 Use case: third-party vendor

There is an IoT data provider that provides IoT sensor data to different companies according to their requirements. The basic objective of the company is to collect and store sensor data and serve this data to the company that demands it. Companies are also using blockchain to store data in the form of smart contracts based on ABAC policy. When any transaction, like saving the IoT data and fetching the IoT data, is requested, the management module ensures the access control policy by referring to the blockchain smart contract.

The process of managing IoT data access leveraging blockchain is illustrated in Fig. 6 which includes IoT data collection and secure data storage in blockchain. In doing this, when a company asks for data, the system checks whether the request is valid and verifies the retention policy of the requested data and, if approved, grants access to the data. We securely deliver the data and immutably log the transaction for transparency and auditing.

3.5 Use case description

The IoT data provider has five types of sensors like LPG, CO, humidity, temperature, and smoke. The Weather Sink Company wants to access humidity and temperature data from a database named IOT_db1; similarly, a public sector company wants to access the LPG, CO, and smoke from a database named IOT_db2. Some users, like U1, U2 and U3, can have access to data related to humidity and temperature, because access rules are based on attributes. Other users like U4, U5, and U6 can have

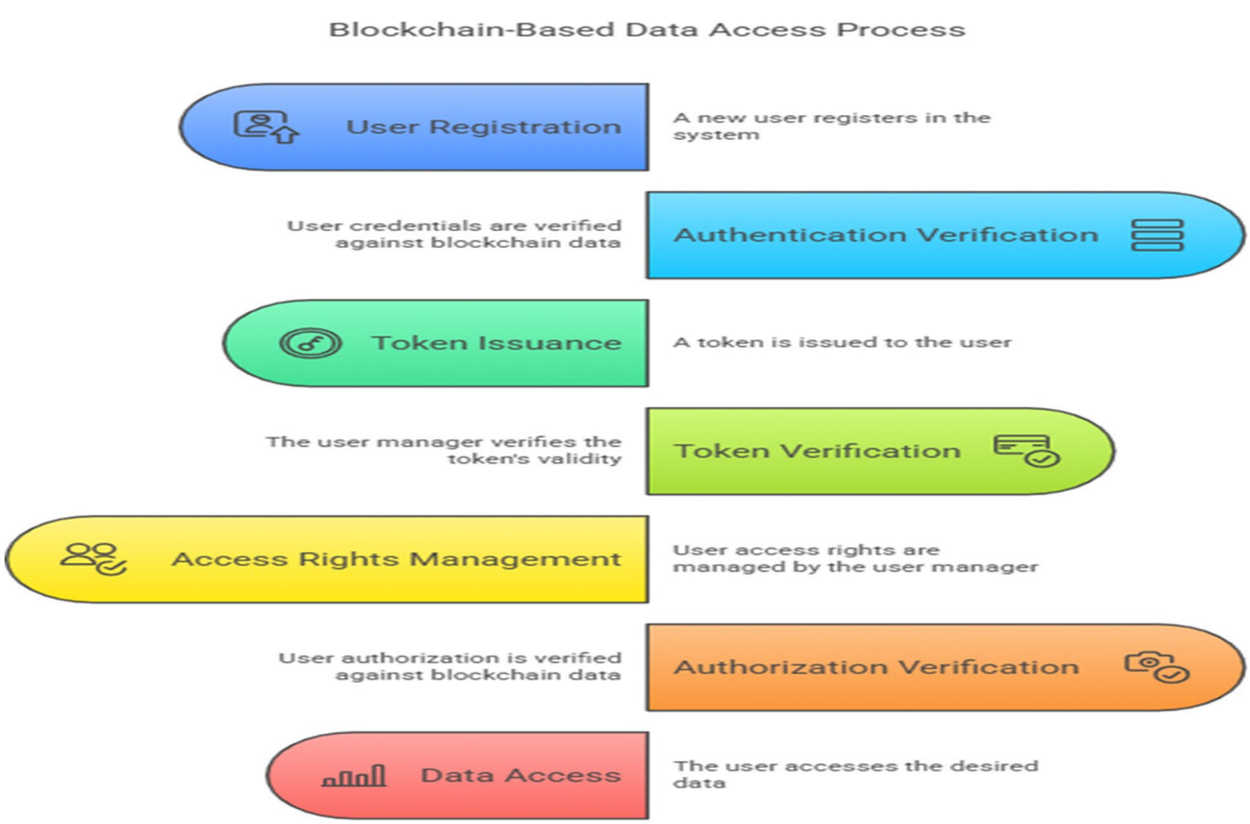

Fig. 5 Blockchain-based data access process

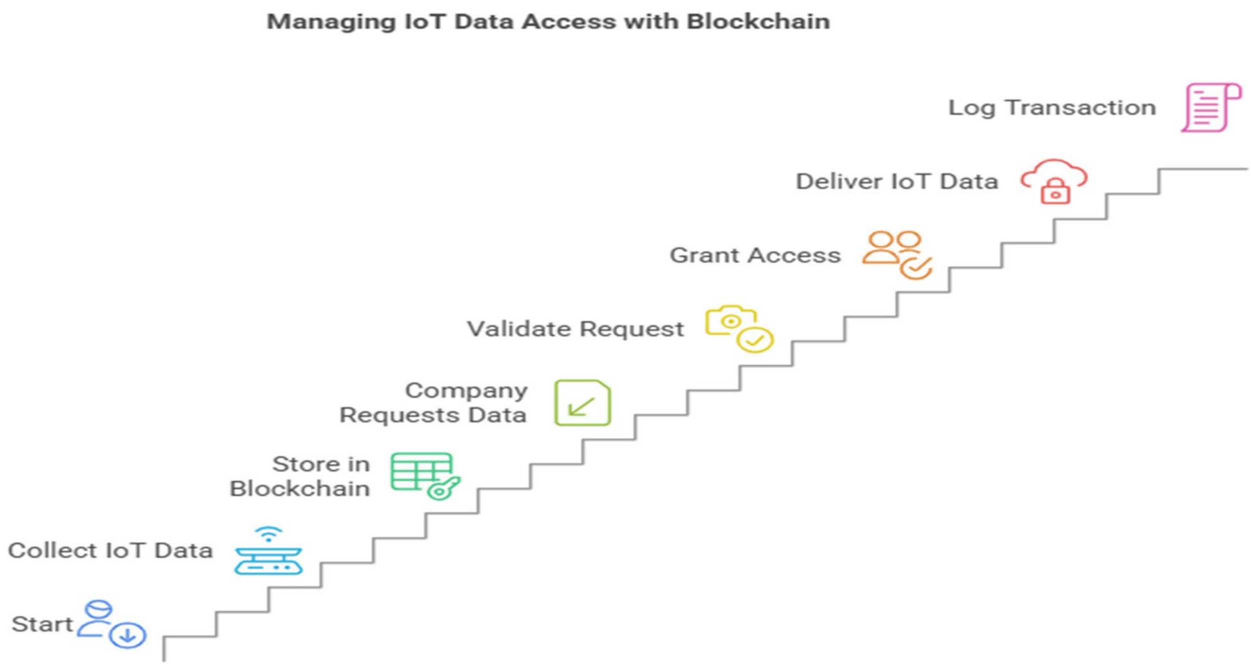

Fig. 6 Managing IoT data access with blockchain

access to public sector companies' data. ABAC policy is used to define the role for users on the basis of attributes.

3.6 Security analysis of the proposed work

The proposed blockchain-based architecture ensures secure storage and retrieval of IoT data by leveraging cryptographic techniques and decentralized consensus mechanisms. Since IoT data credentials and access control policies are stored on the blockchain, they inherit the immutability and tamper resistance of the ledger, preventing unauthorized modifications. Additionally, RabbitMQ, used for scaling blockchain operations, operates with TLS-based encryption to secure data transmission between distributed nodes, mitigating risks of man-in-the-middle attacks. The access control mechanism enforces strict authentication and authorization policies, ensuring that only authorized entities can retrieve or modify IoT data. Furthermore, the blockchain's inherent resistance to single-point failures enhances system reliability, making it resilient to denial-of-service (DoS) attacks. By integrating asynchronous message queuing with blockchain, our approach minimizes the risk of congestion-related attacks, ensuring efficient and secure data flow within the system.

3.7 Security threat modeling analysis

To justify the security of our proposed approach, we perform an extensive security threat modeling analysis to demonstrate its resistance against potential cyber threats. The underlying blockchain level is immutable and stores data integrity and secure access control (IoT credentials and access permissions are protected against unauthorized changes). Yet, blockchain technologies are, though, not immune against Sybil attacks, 51% attacks (in case of public blockchains), or smart contract risks. Our approach tackles these attacks by using a permissioned blockchain scheme, in which known and trustworthy nodes reach consensus, avoiding the possibility that an adversary takes control of the protocol. Moreover, RabbitMQ as message queuing on the basis of RabbitMQ assures the secure and effective communication between IoT equipment and blockchain nodes by distributed processing of high transaction loads, thus resisting denial-of-service (DoS) attack. We also adopt the cryptographic authentication and role-based access control (RBAC) to avert the unauthorized access of the IoT data. For future improvement, a real-time anomaly detection and intrusion detection system will be the pipeline of the proposed security architecture.

3.7.1 Energy and resource trade-offs analysis

The proposed approach balances security, scalability, and resource efficiency by leveraging RabbitMQ for message queuing and blockchain for secure IoT data management. Traditional blockchain-based IoT solutions often suffer from high computational overhead and energy consumption due to frequent consensus operations. However, by offloading transaction processing to RabbitMQ, our method reduces the number of direct blockchain interactions, leading to lower computational costs and network congestion. Additionally, RabbitMQ's asynchronous processing improves resource utilization by handling multiple requests efficiently without overloading blockchain nodes. While blockchain storage introduces higher space requirements, the trade-off is justified by the enhanced security, tamper resistance, and decentralized access control it provides. This balance ensures that the proposed system remains scalable and energy-efficient, making it a practical solution for real-world IoT deployments.

Forgiving the decentralized and scalable architecture of our framework, its key limitations are as follows: (1) static smart contract-based ABAC policies, which need a smart contract update to be modified, (2) the latency of blockchain consensus carrying the risk of slow access control decisions in time-sensitive IoT conflict, and (3) the fault-tolerant asynchronous message queuing approach may introduce additional delays in frequent access requests. In conclusion, such limitations offer vital guidelines that could be taken into consideration for the development of more adaptive policy mechanisms, not to mention enabling to strike better balance between security, scalability, and real-time performance in IoT ecosystems.

4 Implementation

4.1 Development environment

System specification for the proposed work is as follows: Windows 10 Pro with an Intel® Core™ i7-2000U CPU at 2.50 GHz, 16 GB RAM, and a 64-bit operating system; Python programming is chosen for coding, running the program on Jupyter Notebook 7.0.3, and storing the data in the MongoDB database. MongoDB is a document database. Studio3T is used as a MongoDB IDE, because it provides a GUI that has auto-completion and system highlighting, built-in JSON validation, and automatic query code generation and works faster and saves time.

In order to execute the message queuing system, RabbitMQ 3.12.2 is used in our work, and to run RabbitMQ, Erlang language is required. So, Erlang 26.0.2 is installed.

A smart contract is written in the form of an ABAC policy and stored on the blockchain server. User requests are sent to the blockchain server through RabbitMQ, which is a message broker. There are two ways in which message brokers communicate: one is the point-to-point model, and the other is the producer–consumer or publish-subscribe model for communication between IoT gadgets, blockchain network, and users.

4.2 Blockchain-based attribute-based access control (ABAC): secure, decentralized, and auditable access control implementation

ABAC policies can be validated through the MRA model via the following three layers: an ABAC policy layer is set to define who can access which resources under what conditions and is stored on the blockchain along with the resources. This request is logged as a transaction where the identity provider authenticates the user. Then, the smart contract checks if Alice satisfies the policy conditions like {required_role: HR Manager, min_clearance: 2}, and if she does, it grants her access. If not, the request is denied. The data is stored securely, and all access decisions are recorded immutably on the blockchain with authorized auditors capable of tracing transactions, confirming policy compliance and finding anomalies. This system, capable of operating in real time while providing tamper-proof access control, adds a new level of security, transparency, and scalability across different domains, enabling seamless, trustless authorization in decentralized ecosystems.

In Fig. 7, the structure of the used blockchain-based ABAC with RabbitMQ system is presented, whereas the main aspects of the system include system setup (installation of Erlang and RabbitMQ), user request handling (transmission of requests through RabbitMQ), smart contract (where the ABAC policy is enforced), authentication and policy validation (user verification and policy checking), access decision (granting or denying access), and logging and auditing (keeping the decisions on the blockchain).

5 Result

5.1 Results and discussion

For experimental evaluation, we employed public IoT weather data with a time-series excerpt of around 12 IoT sensors gathered within the last 2 years; we used this data to simulate realistic IoT access control and sensor interaction qualitative attacks. We also used synthetic datasets to simulate different ABAC policies with various roles and access operations, to examine the performance of our framework under different access control situations and diverse policy complexities. The mix of synthetic ABAC scenarios and real IoT sensor data for testing our blockchain-RabbitMQ integration is quite complementary, which enables us to cover a full set of our testing with fine-granularity dataset, simulation parameter, and access control policy configuration details.

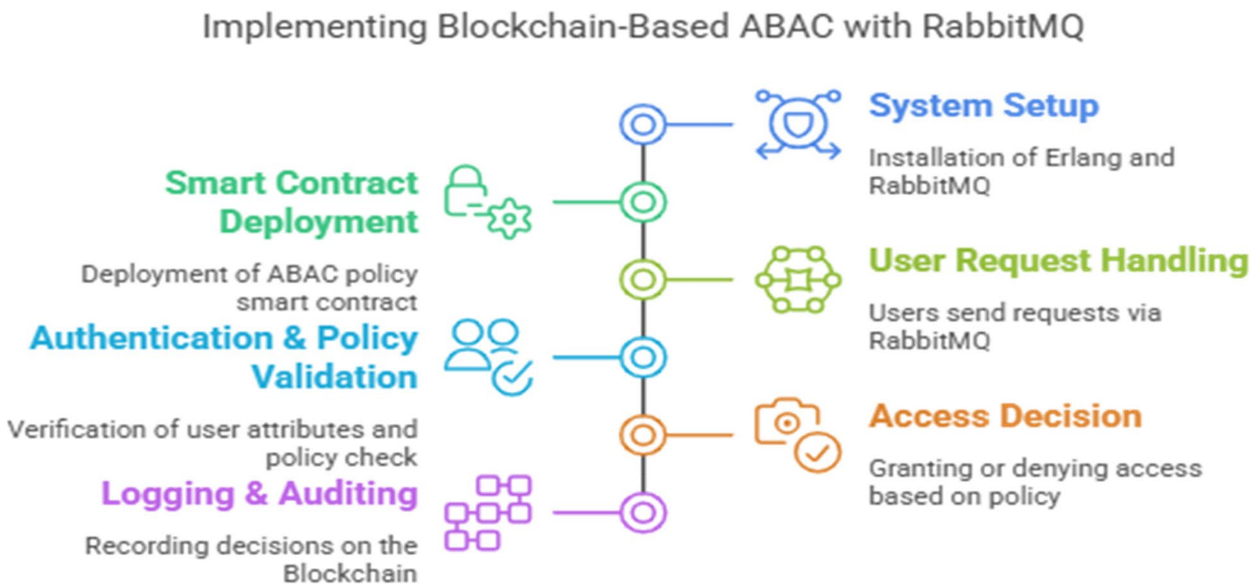

Fig. 7 Implementing blockchain-based ABAC with RabbitMQ

5.2 Performance analysis of the producer–consumer paradigm

It examined the producer consumer model in blockchain transactions. In the experimental model, producers acted in the role of message publishers and consumers in the role of transaction processors. Our results suggest transaction volume exceeds some threshold (e.g., 1000) where the number of consumers needs to increase in order to keep the system efficient.

The obtained results support the claim that by increasing the consumer number, the time spent on executing transactions is reduced, and the system scalability is improved as the load of processing the transactions is balanced. But according to what we know from multiprocessing, the parallel execution will decrease the finish time, but the improvement would be sub-linear, as the overhead was added from context switching and resource contention.

5.3 Execution time analysis

Table 2 presents the relationship between message volume, consumer count, and execution time. For a constant message volume, execution time decreases as consumer count increases, though not proportionally, due to coordination overhead (Fig. 8).

For 2000 messages, increasing consumers from 2 to 4 reduced execution time by approximately 49.7%, while further doubling from 4 to 8 yielded only a 48.1% reduction, demonstrating diminishing returns with additional consumers. Similar patterns were observed with 4000 messages, confirming that the architecture scales effectively but with expected overhead limitations (Fig. 9).

The system successfully processed 5000 transactions without performance degradation in terms of CPU utilization, memory consumption, or memory leakage, validating the architecture's scalability potential.

Table 2 Execution time performance analysis

No. of Messages	No. of Consumers	Execution Time (hr:min:sec)
2000	2	1:24:10
2000	4	0:42:24
2000	6	0:28:01
2000	8	0:22:00
2000	10	0:17:03
4000	2	3:28:00
4000	4	1:23:00
4000	6	0:56:00
4000	8	0:42:00
4000	10	0:34:00

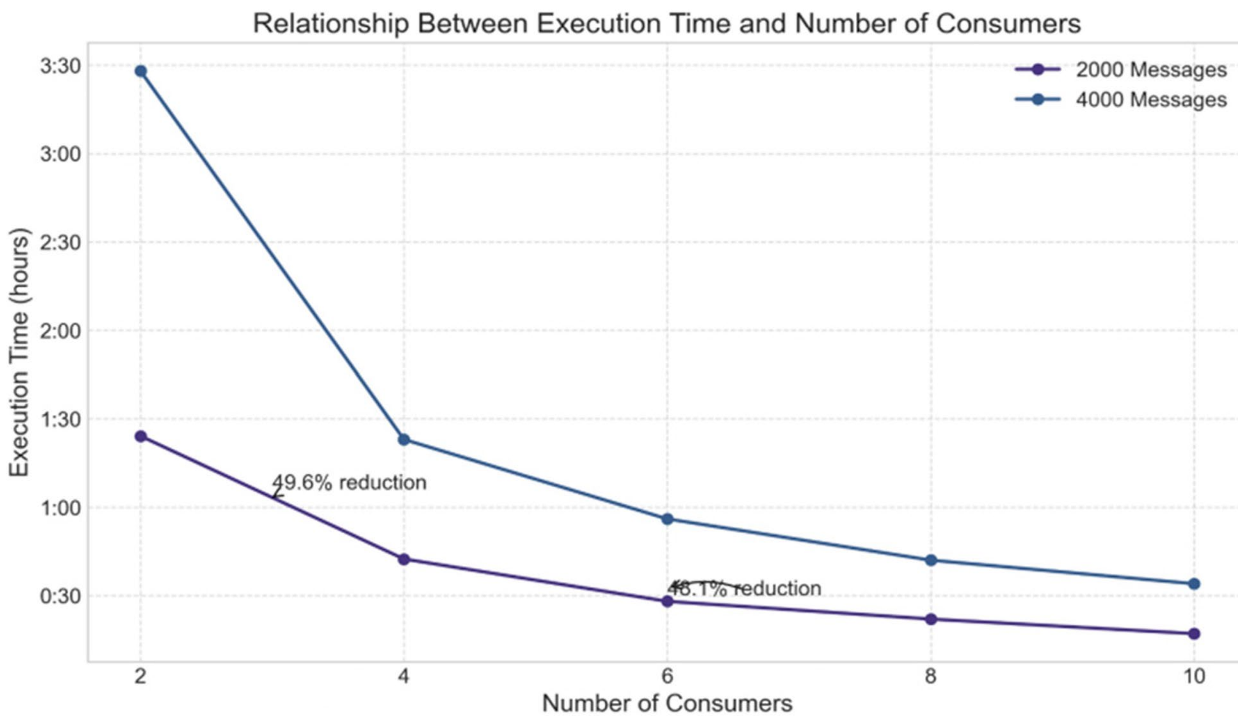

Fig. 8 The relationship between execution time and number of consumers shows a nonlinear improvement pattern, demonstrating diminishing returns as more consumers are added

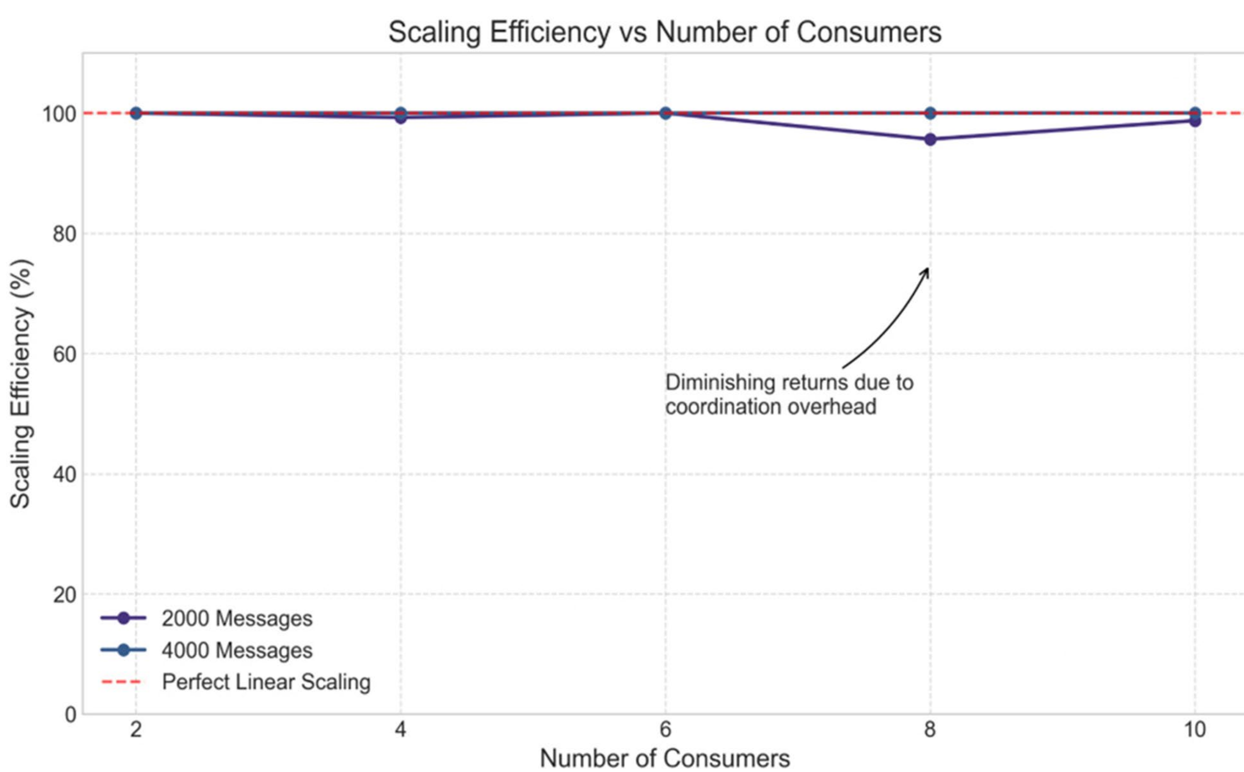

Fig. 9 Scaling efficiency decreases as consumer count increases, illustrating the diminishing returns phenomenon caused by coordination overhead

Table 3 Process memory utilization analysis

No. of Messages	No. of Consumers	Process Memory (MiB)
2000.0	2.0	2.8
2000.0	4.0	2.8
2000.0	6.0	1.3
2000.0	8.0	2.8
2000.0	10.0	4.5
4000.0	2.0	6.4
4000.0	4.0	11.0
4000.0	6.0	7.3
4000.0	8.0	11.0
4000.0	10.0	6.4

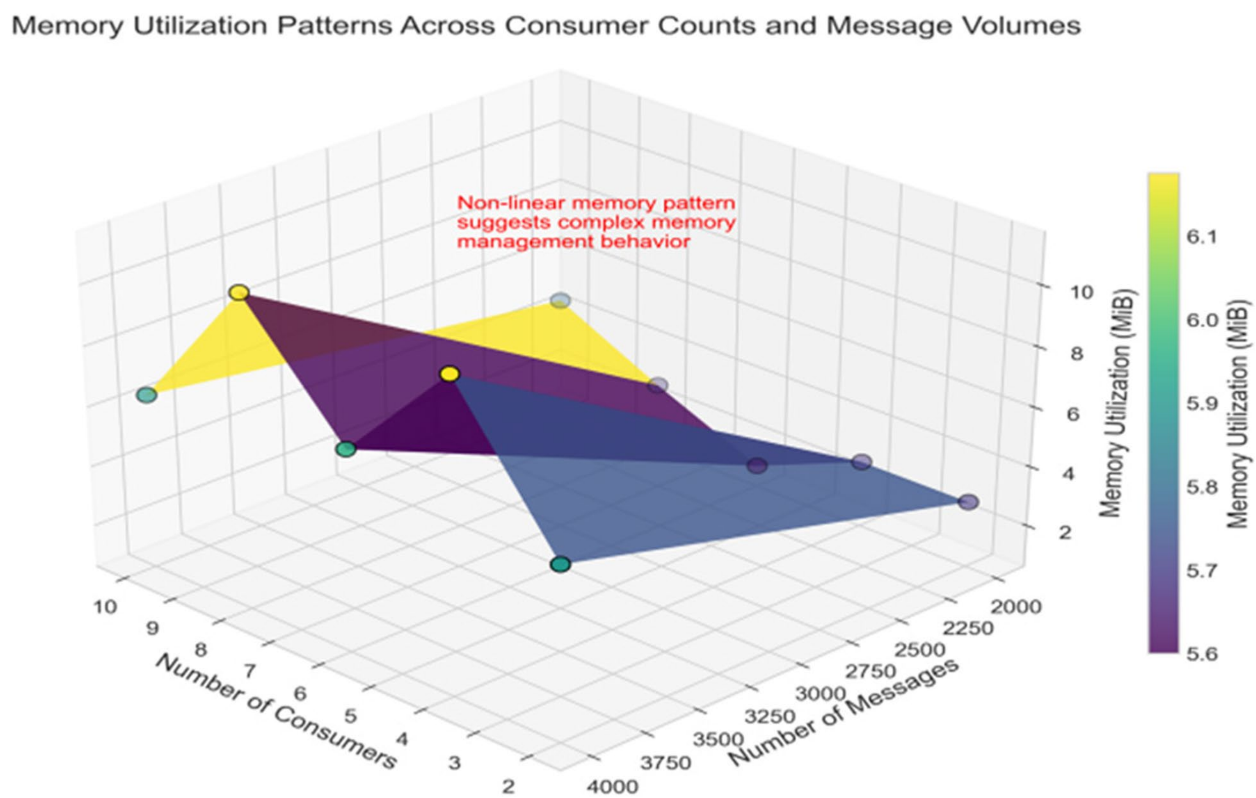

Fig. 10 Memory utilization exhibits nonlinear patterns across different consumer counts, suggesting complex memory management behavior in the system

5.4 Memory utilization analysis

Memory utilization was monitored across varying message volumes and consumer counts to evaluate resource efficiency. As shown in Table 3, memory consumption patterns demonstrate complex behavior that does not strictly correlate with increases in consumer count (Fig. 10).

The nonlinear relationship between consumer count and memory utilization can be attributed to memory allocation strategies, garbage collection cycles, and process management overhead within the implementation.

Table 4 Acknowledgement delivery analysis

No. of Messages	No. of Consumers	Deliver Ack. (per second)
2000	2	140
2000	4	65
2000	6	40
2000	8	50
2000	10	40
4000	2	350
4000	4	100
4000	6	90
4000	8	80
4000	10	45

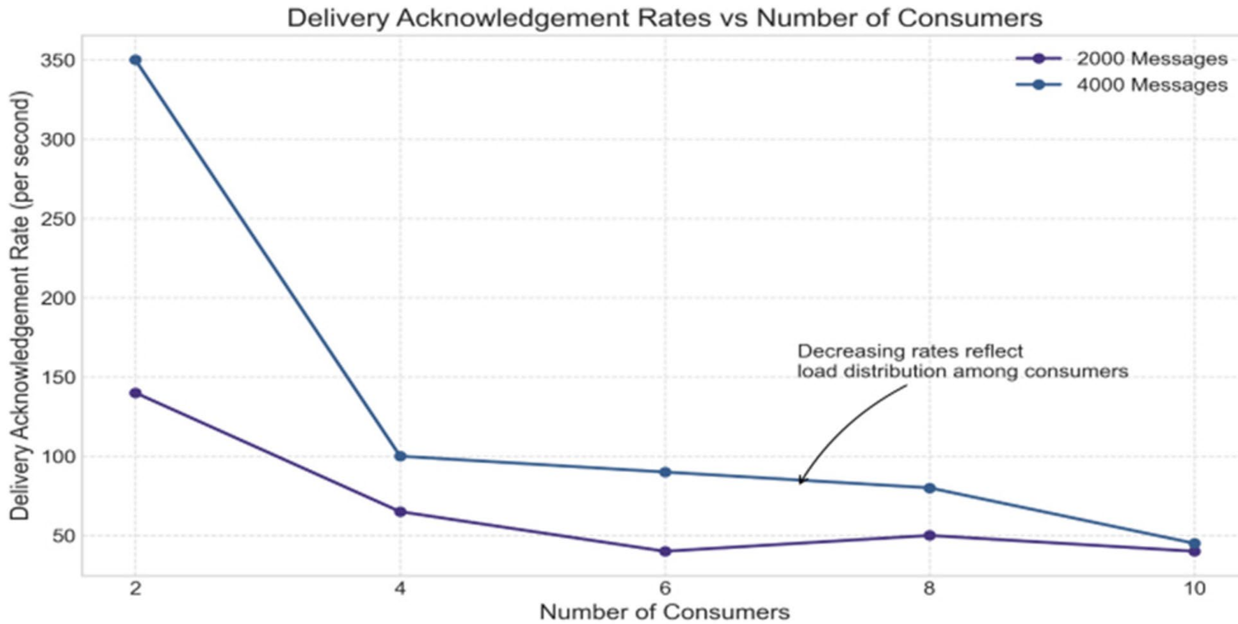

Fig. 11 Delivery acknowledgement rates decrease as consumer count increases, reflecting load distribution among consumers

Table 5 Consumer acknowledgement analysis

No. of Messages	No. of Consumers	Msg Rate (Consumer Acknowledgement) (per Sec
2000.0	2.0	0.4
2000.0	4.0	0.8
2000.0	6.0	1.2
2000.0	8.0	1.6
2000.0	10.0	2.0
4000.0	2.0	0.4
4000.0	4.0	0.8
4000.0	6.0	1.2
4000.0	8.0	1.6
4000.0	10.0	2.0

Memory consumption did not increase proportionally with message volume, suggesting efficient memory management in the architecture.

5.5 Message acknowledgement analysis

Delivery acknowledgement rates were measured to evaluate system throughput under varying loads. Table 4 demonstrates that as consumer count increases, the per-second acknowledgement delivery rate generally decreases, indicating load distribution among multiple consumers (Fig. 11).

Conversely, consumer acknowledgement rates (Table 5) demonstrate a linear relationship with consumer count, independent of message volume. This indicates that individual consumer processing capacity remains consistent regardless of total system load (Fig. 12).

5.6 Blockchain network performance

The blockchain network was implemented using Python, with nodes communicating through the producer–consumer paradigm. System resource utilization was measured to evaluate network performance under varying loads.

5.6.1 System resource utilization

Table 6 presents the CPU and memory utilization statistics across different consumers. The results demonstrate that increasing the number of consumers has minimal impact on the system's resource utilization parameters (Figs. 13 and 14).

This stability in resource utilization metrics across varying consumer counts validates that the proposed architecture efficiently manages resources without imposing additional system burden when scaling horizontally. The minimal variations observed in CPU and memory utilization suggest that the blockchain implementation effectively balances computational load across available resources.

5.7 Security analysis

The implementation was evaluated against key security requirements for distributed systems. The architecture successfully addresses the following:

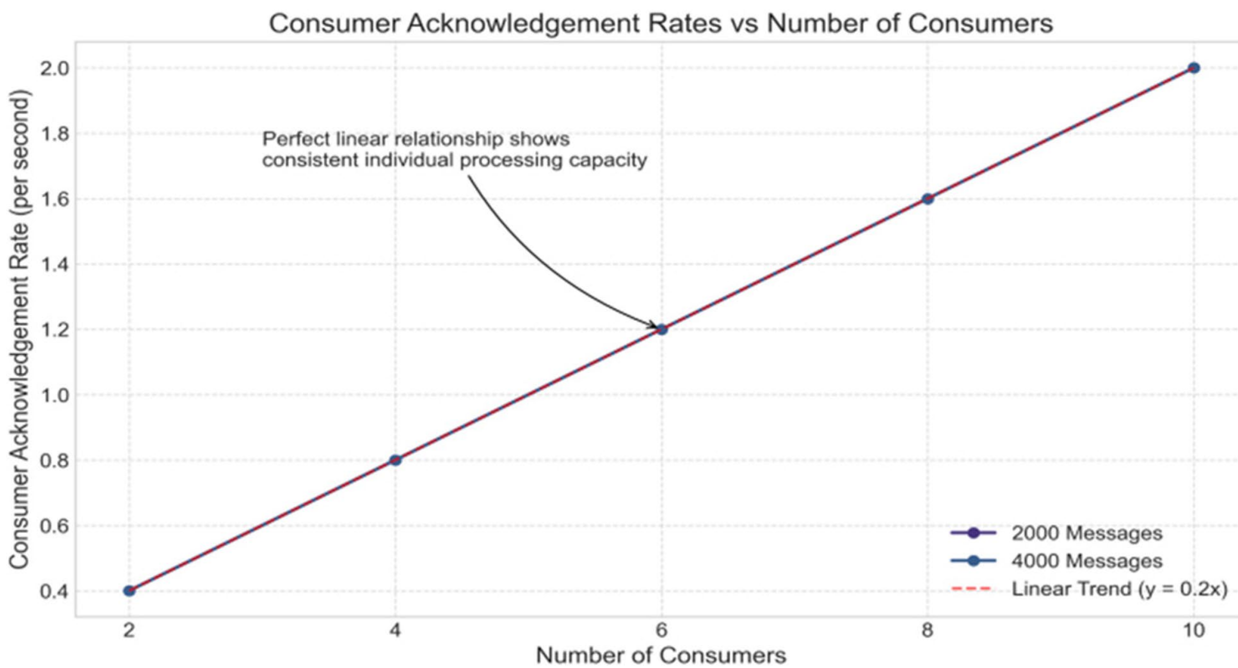

Fig. 12 Consumer acknowledgement rates show a linear relationship with consumer count, demonstrating consistent individual processing capacity

Table 6 Network performance analysis

No. of Consumers	CPU MIN	CPU MAX	CPU STD	MEM MIN	MEM MAX	MEM STD
2.0	33.2	81.9	16.95	49.8	50.2	0.1
4.0	33.0	81.9	16.0	49.8	50.2	0.1
6.0	33.0	81.9	16.0	49.8	50.2	0.1
8.0	34.0	85.2	19.0	49.4	49.9	0.14
10.0	35.4	93.0	21.8	49.5	49.8	0.11

1. *Confidentiality*: Access control is implemented through token-based authentication, restricting system access to authorized users with valid tokens.
2. *Integrity*: Data integrity is maintained through the inherent properties of blockchain technology. Each block incorporates the hash value of its predecessor, making unauthorized modifications readily detectable.
3. *Availability*: System resources and data are consistently available to authenticated users.
4. *Authentication*: The token-based system verifies user identity before granting system access.
5. *Authorization*: Access to specific data is governed by attribute-based access control (ABAC) policies implemented as smart contracts, ensuring users can only access permitted resources (Fig. 15).

5.8 Discussion

The experimental results demonstrate that the proposed blockchain architecture efficiently handles transaction processing through horizontal scaling. The sublinear performance improvement observed when increasing consumer count aligns with established distributed systems theory, where coordination overhead and resource contention limit perfect linear scaling (Fig. 16).

The relatively stable memory utilization across configurations suggests efficient memory management

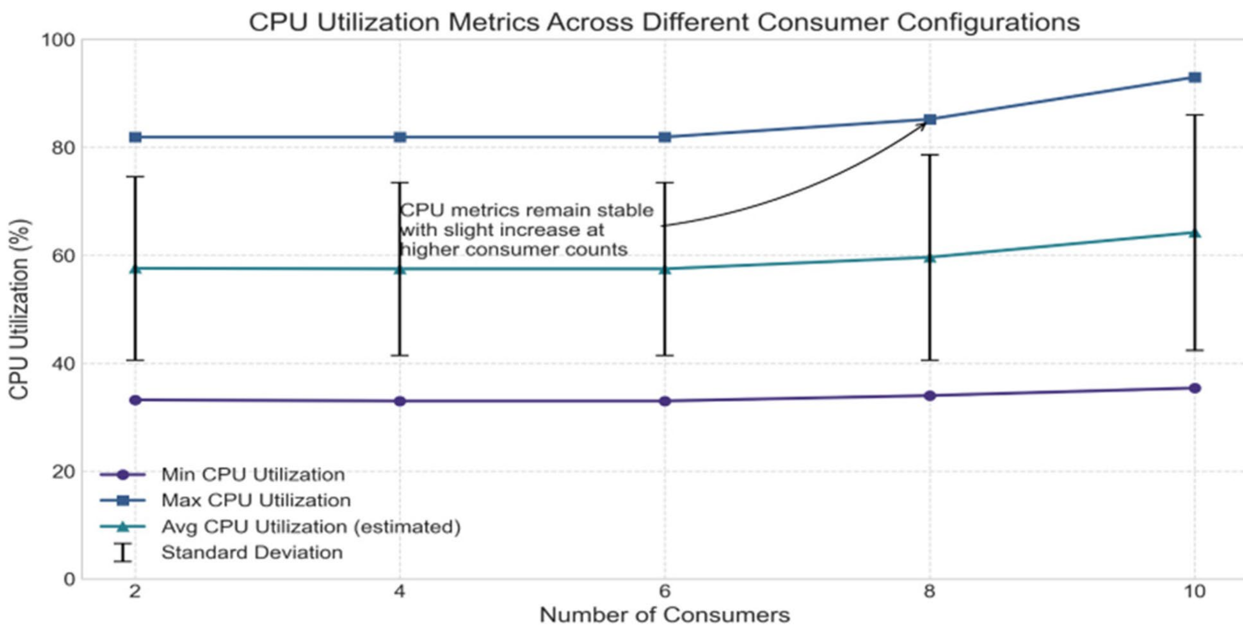

Fig. 13 CPU utilization metrics remain relatively stable across different consumer configurations, with slight increases at higher consumer counts

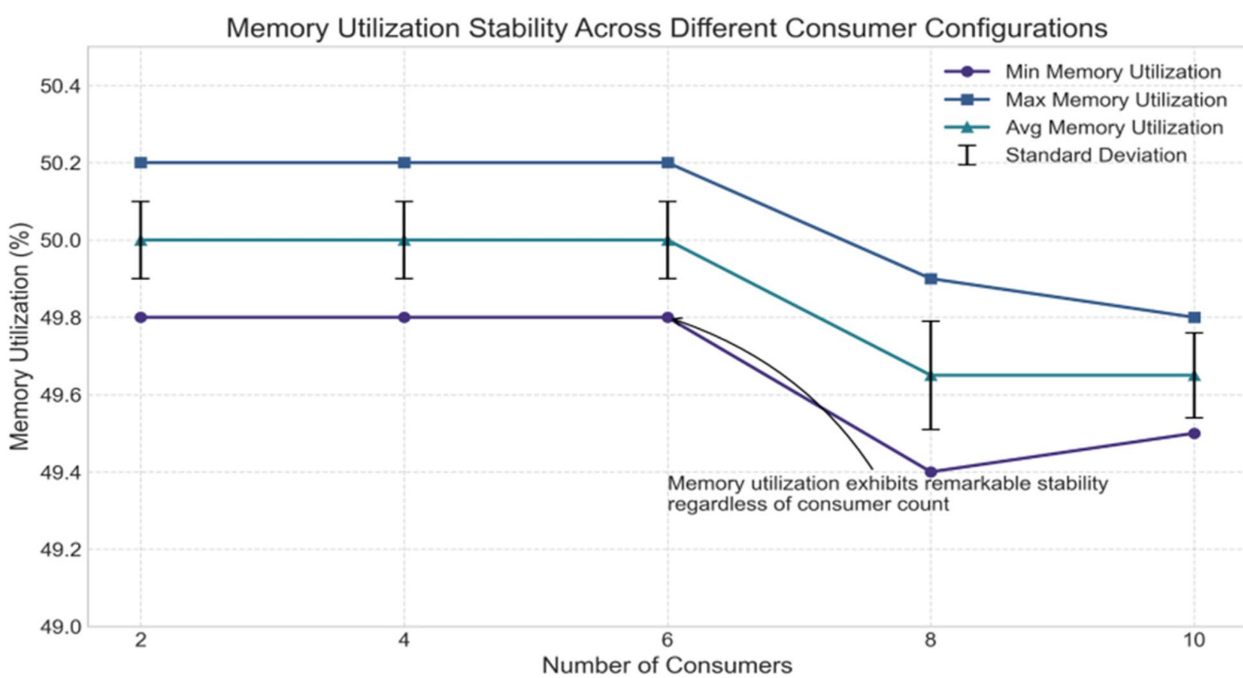

Fig. 14 Memory utilization exhibits remarkable stability regardless of consumer count, suggesting efficient memory management in the system

within the implementation. Similarly, the consistent system resource utilization metrics indicate that the architecture efficiently manages computational resources even as consumer count increases.

5.9 Block creation time performance

Figure 17 illustrates the block creation time comparison between traditional blockchain-based access control systems and our proposed framework. The results demonstrate significant improvements in block creation

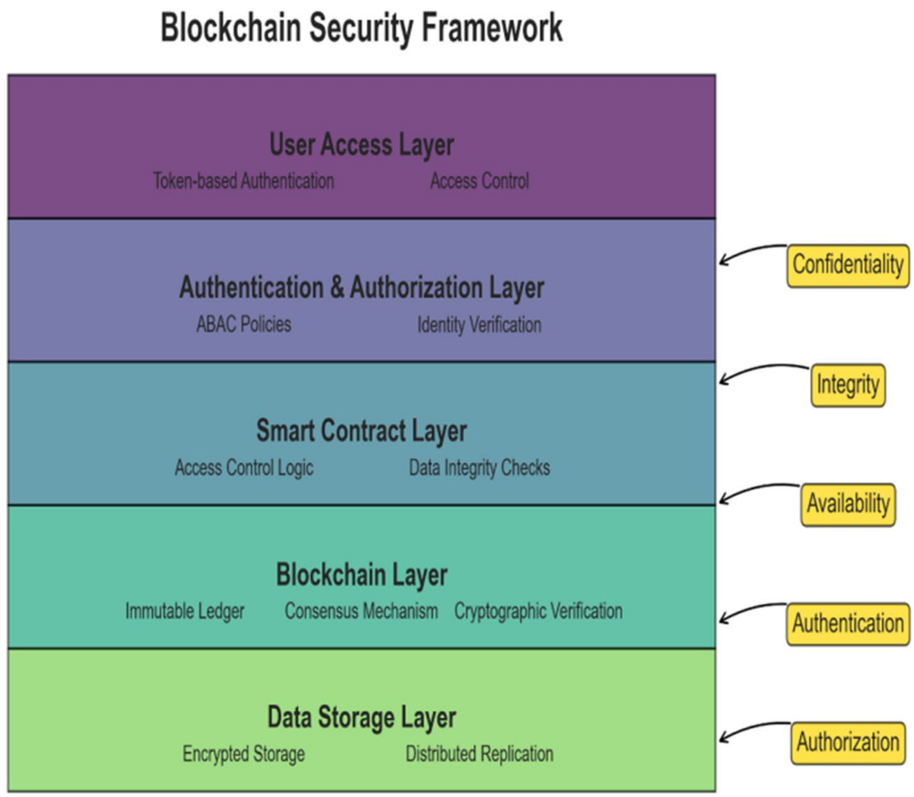

Fig. 15 The security framework implements multiple layers of protection, addressing key security requirements for distributed blockchain systems

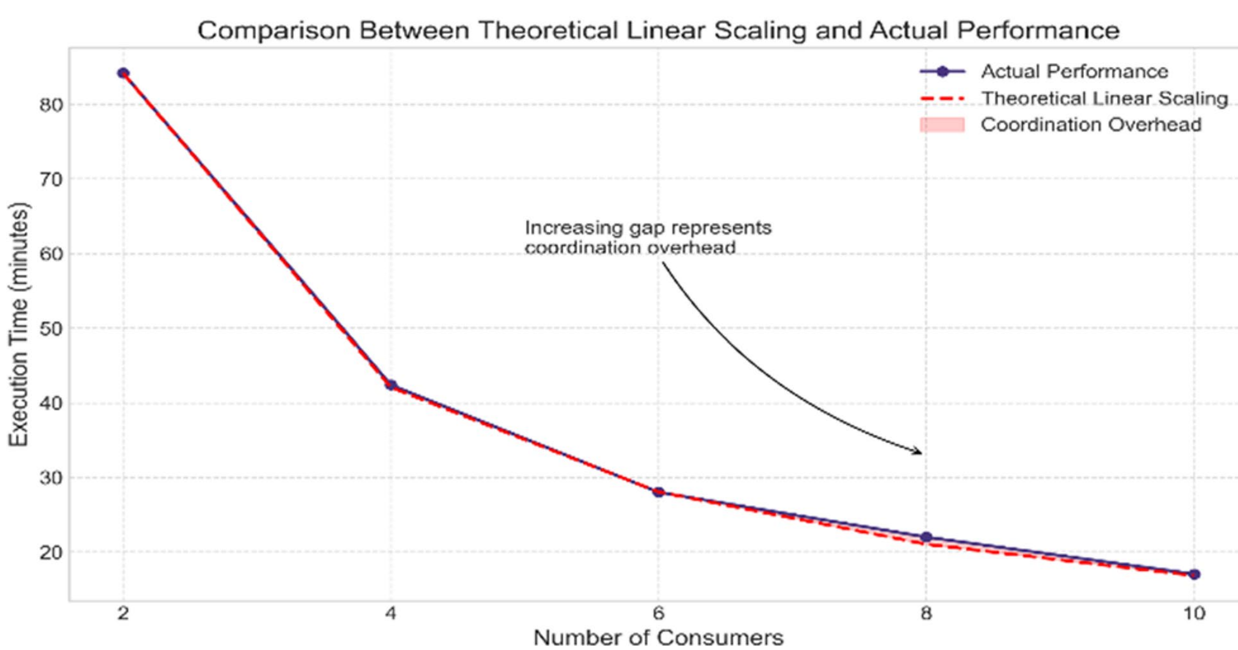

Fig. 16 Comparison between theoretical linear scaling and actual measured performance illustrates the impact of coordination overhead on system scalability

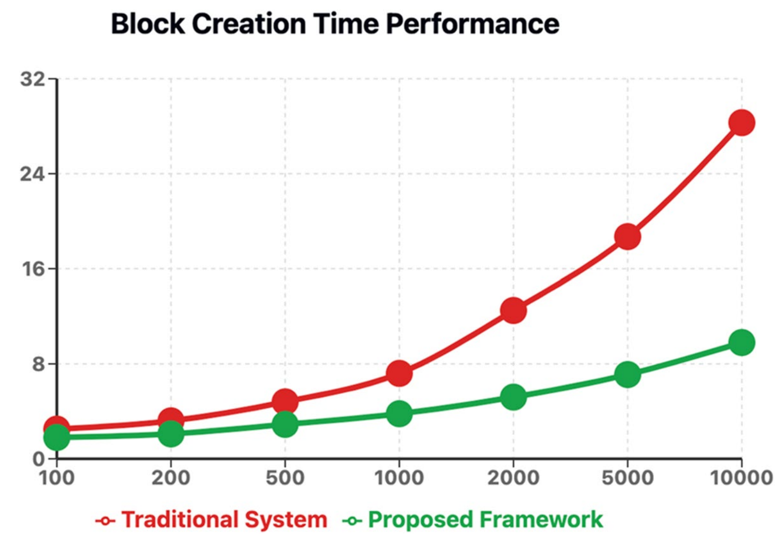

Fig. 17 Comparison of block creation time between traditional system and proposed framework. *X*-axis, number of IoT devices; *Y*-axis, block creation time (s)

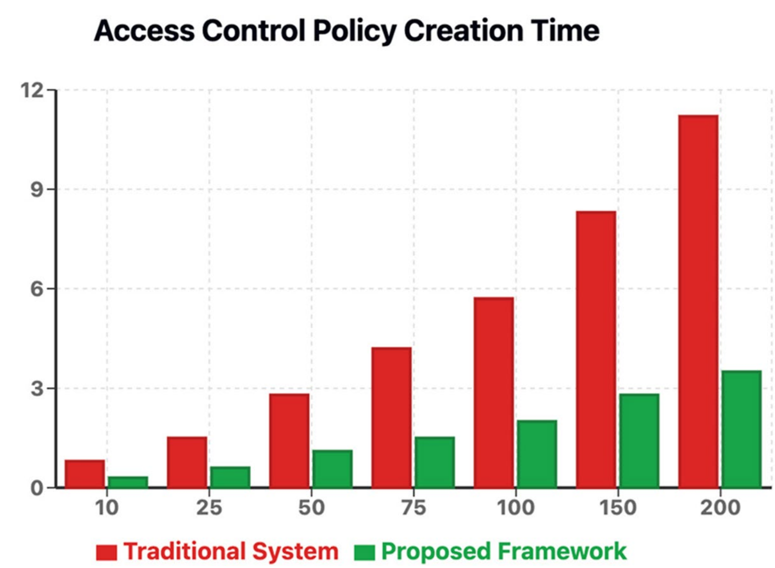

Fig. 18 Comparison of access control policy creation time between traditional system and proposed framework. *X*-axis, number of ABAC policies; *Y*-axis, policy creation time (s)

efficiency, particularly as the number of IoT devices increases.

5.10 Access control policy creation time

Figure 18 presents the access control policy creation time analysis, showing how our smart contract-based ABAC implementation outperforms traditional policy management systems in terms of policy deployment and execution efficiency.

The security implementation addresses the core requirements of confidentiality, integrity, availability, authentication, and authorization through a combination of blockchain's inherent security properties and additional access control mechanisms.

These findings confirm that the proposed architecture for transaction processing systems can grow and is secure, with possible uses in distributed ledger technologies that need to handle a lot of transactions and provide strong security.

Our study also reveals that our approach succeeds in protecting RabbitMQ against DDoS attacks using the distributed storage and queueing mechanisms, and that the decentralized architecture based on the blockchain technology (and the involvement of multiple nodes) makes them less vulnerable to 51% attacks by the need for consensus among all the nodes in the network. In the case of phishing attacks, our smart contract-driven ABAC policies allow strict cryptographic assurance of device identity and attributes so unauthorized access can be avoided even if authentication is compromised. We have also shown that the framework can protect against replay, man-in-the-middle, and Sybil attacks, and that using blockchain's unchangeable records, message queue authentication, and detailed access control rules provides complete security in the IoT area.

The complexities addressed by our framework include the following: (1) Device heterogeneity in terms of varying computational capabilities, communication protocols, and security requirements across different IoT devices, (2) dynamic trust levels where device trustworthiness may change based on behavior patterns and environmental conditions, and (3) scalability challenges arising from the massive number of devices generating high-volume transactions that require efficient access control decisions. We have clarified these specific complexities in the introduction and related work sections to better contextualize how our blockchain-RabbitMQ integration and ABAC implementation address these particular challenges in IoT environments.

6 Conclusion and future work

In this experiment, we discovered the integration of IoT control of access systems with blockchain technology and message queuing systems to enhance security, transparency, and efficiency. Our proposed system leverages blockchain's immutability and decentralized nature to secure control of access records and improve reliability. Additionally, message queuing systems facilitate efficient and reliable communication between IoT devices, ensuring appropriate and accurate control of access operations.

The implementation demonstrated several key advantages as follows:

- *Enhanced security*: Blockchain's cryptographic mechanisms safeguard control of access data against tampering and unauthorized alterations.
- *Increased transparency*: The absolute record keeper provided by blockchain ensures that all access attempts are recorded and traceable.
- *Improved efficiency*: The message queuing system optimizes communication and processing speeds between IoT gadgets, reducing latency and potential restricted access.

The integration of these technologies addresses many of the vulnerabilities and limitations inherent in traditional IoT control of access systems, setting a new standard for secure and efficient access management.

Despite the promising results, future work could further improve the system. Future studies should focus on scaling solutions, such as layer-2 protocols or alternative consensus techniques, maintaining performance as the system grows, and exploring standards and frameworks to ensure interoperability between different IoT gadgets, blockchain platforms, message queuing systems, and more energy-efficient consensus algorithms and techniques to reduce the environmental impact of the system.

Authors' contributions

N.K. has done the research and written the paper,A.S. is my P.hD guide,she has guided me through out the work.

Funding

No funding.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

There is no informed consent for this study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 16 March 2025 Accepted: 26 June 2025
Published online: 24 October 2025

References

1. Y. Liu et al., Capability-based IoT access control using blockchain. Digit. Commu. Netw. **7**, 463–469 (2021). https://doi.org/10.1016/j.dcan.2020.10.004
2. C. Cheng, et al., The blockchain based access control scheme for the Internet of Things. International Conference on Identification Information and Knowledge in the Internet of Things. (2021)
3. Adam Ibrahim Abdi, et al. Hierarchical blockchain-based multi-chaincode access control for securing IoT system. Electronics 2022, 11,711.https://doi.org/10.3390/electronics11050711
4. H. Liu, D. Han, D. Li, Fabric-IoT: a blockchain-based access control system in IoT. IEEE Access **8**, 18207–18218 (2020). https://doi.org/10.1109/ACCESS.2020.2968492
5. W. Liang, M. Tang, J. Long, X. Peng, J. Xu, K. Li, A secure fabric blockchain-based data transmission technique for industrial Internet-of-Things. IEEE Trans. Industr. Inf. **15**(6), 3582–3592 (2019). https://doi.org/10.1109/TII.2019.2907092

6. H. Li, D. Han, EduRSS: a blockchain-based educational records secure storage and sharing scheme. IEEE Access **7**, 179273–179289 (2019). https://doi.org/10.1109/ACCESS.2019.2956157
7. Y.E. Oktian, S.G. Lee, BorderChain: blockchain-based access control framework for the Internet of Things endpoint. IEEE ACCESS **9** (2021). https://doi.org/10.1109/ACCESS.2020.3047413
8. S. Biswas, K. Shaif, et al., A scalable blockchain framework for secure transaction in IoT. IEEE Internet Things J. https://doi.org/10.1109/JIoT.2018.2874095.
9. V. Agarwal et al., Blockchain meets IoT: a scalable architecture for security and maintenance. IEEE 17th International Conference on Mobile Ad hoc and Sensor Systems(MASS). https://doi.org/10.1109/MASS50613.2020.00017.
10. R, Fotohi, et al., Securing communication between things using blockchain technology based on authentication and SHA-256 to improving scalbility in large scale IoT. Computer networks 197, 1389–1286/@ elsevier (2021). https://doi.org/10.1016/j.comnet.2021.108331
11. S. Chaisawat, C. Vorakulpipat, Towards achieving personal privacy protection and data security on integrated e-voting model of blockchain and message queue. Secur. Commun. Netw. **2021**, 1–14 (2021). https://doi.org/10.1155/2021/8338616
12. S. Sun, et. al., Blockchain-based IoT access control system: towards security, lightweight, and cross-domain. IEEE ACCESS, Vol. 9, (2021). https://doi.org/10.1109/ACCESS.2021.3059863.
13. S. Tanwar, et al., Next generation IoT and blockchain integratioN. Hindawi Journal of Sensors, vol. 2022, paper ID 9077348, 14. https://doi.org/10.1155/2022/9077348
14. A. AL. Sadawi, et al., A survey on the integration of blockchain with IoT to enhance performance and eliminate challenges. IEEE ACCESS, Open access journal, Vol. 9, (2021). https://doi.org/10.1109/ACCESS.2021.3070555
15. S. singh, et al., SH-BlockCC: a secure and efficient Internet of things smart home architecture based on cloud computing and blockchain technology. Int. J. Distributed Sensor Networks 2019, vol.15(4)© The Author(s) (2019). https://doi.org/10.1177/1550147719844159
16. O.A. Khashan et al., Efficient hybrid centralized and blockchain-based authentication architecture for heterogeneous IoT systems. J. King Saud Uni. Compu. Inf. Sci **35**, 726–739 (2023). https://doi.org/10.1016/j.jksuci.2023.01.011
17. R. Raman, M. A. Verma, V. Kumar, S. Rastogi, B. G. Pillai and R. Meenakshi, Fog computing integrated with and blockchain technology for accurate disease prediction. 2024 4th International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE), Greater Noida, India, 2024, pp. 677–682. https://doi.org/10.1109/ICACITE60783.2024.10616443
18. S. A. Z. Hassan, B. E. Elakhdar, W. M. Saied and D. G. Hassan, Leveraging new technologies for building a comprehensive smart MIS: integrating ERP, blockchain, IoT, context-awareness, and cloud computing. 2024 6th International Conference on Computing and Informatics (ICCI), New Cairo - Cairo, Egypt, 2024, pp. 459–465. https://doi.org/10.1109/ICCI61671.2024.10485102
19. J. Lee, J. Oh, D. Kwon, M. Kim, K. Kim, Y. Park, Blockchain-enabled key aggregate searchable encryption scheme for personal health record sharing with multidelegation. IEEE Internet Things J. **11**(10), 17482–17494 (2024). https://doi.org/10.1109/JIOT.2024.3357802
20. H. Kumazaki and O. Mizuno, Evaluation of applying blockchain technology to IoT data distribution. 2023 28th Asia Pacific Conference on Communications (APCC), Sydney, Australia, 2023, pp. 318–323. https://doi.org/10.1109/APCC60132.2023.10460717
21. I. BOURIAN, A. SEBBAR, K. CHOUGDALI and E. M. Amhoud, SSHCEth: Secure Smart Home Communications based on Ethereum Blockchain and Smart Contract. GLOBECOM 2023 - 2023 IEEE Global Communications Conference, Kuala Lumpur, Malaysia, 2023, pp. 2674–2679. https://doi.org/10.1109/GLOBECOM54140.2023.10437193
22. H. Mhamdi, A. Zouinkhi and M. N. Abdelkrim, "A brief overview of blockchain technology and access control in IoT," 2023 IEEE International Workshop on Mechatronic Systems Supervision (IW_MSS), Hammamet, Tunisia, 2023, pp. 1–5. https://doi.org/10.1109/IW_MSS59200.2023.10369788
23. D.K. Murala, S.K. Panda, S.P. Dash, MedMetaverse: medical care of chronic disease patients and managing data using artificial intelligence, blockchain, and wearable devices state-of-the-art methodology. IEEE Access **11**, 138954–138985 (2023). https://doi.org/10.1109/ACCESS.2023.3340791
24. R. Maruthi, D. Piriadarshani, S. Padmanabhan and M. Shanthi, "A secured framework for blockchain technology adoption in IoT," 2023 2nd International Conference on Edge Computing and Applications (ICECAA), Namakkal, India, 2023, pp. 1366–1369. https://doi.org/10.1109/ICECAA58104.2023.10212185
25. A. Sadad et al., Distributed twins in edge computing: blockchain and IOTA. 2023 International Wireless Communications and Mobile Computing (IWCMC), Marrakesh, Morocco, 327–332 (2023). https://doi.org/10.1109/IWCMC58020.2023.10182509
26. I. Singh and B. Singh, Integration of decentralized blockchain with Cloud & IoT based SCM. 2023 International Conference on Advancement in Computation & Computer Technologies (InCACCT), Gharuan, India, 887–892 (2023). https://doi.org/10.1109/InCACCT57535.2023.10141797
27. H. Li, D. Han, C.C. Chang, DAC4SH: a novel data access control scheme for smart home using smart contracts. In IEEE Sensors J. **23**(6), 6178–6191 (2023). https://doi.org/10.1109/JSEN.2023.3241093
28. Y. Guo, Z. Lu, H. Ge, J. Li, Revocable blockchain-aided attribute-based encryption with escrow-free in cloud storage. IEEE Transactions on Computers **72**(7), 1901–1912 (2023). https://doi.org/10.1109/TC.2023.3234210
29. A. Alanazi, A. Alqahtani, S. Alsubai and M. Bhatia, IoT-inspired smart theft control framework for logistic industry, IEEE Internet Things J. https://doi.org/10.1109/JIOT.2024.3445884
30. O. Ishmilh, M. A. Jarwar, Y. Javed, A novel IoT middleware for secure pharmaceuticals condition monitoring in supply chain. 2024 Fifteenth International Conference on Ubiquitous and Future Networks (ICUFN), Budapest, Hungary, 342–347 (2024). https://doi.org/10.1109/ICUFN61752.2024.10625066
31. T. Xu et al., Cyclic sensing: an orbital spectrum sensing method for uplink NOMA IoT systems. ICC 2024 - IEEE International Conference on Communications, Denver, CO, USA, 373–378 (2024). https://doi.org/10.1109/ICC51166.2024.10622988
32. V. Kavididevi, S. K. Monikapreethi, M. Rajapriya., P. S. Juliet, S. Yuvaraj, M. Muthulekshmi, IoT-enabled reinforcement learning for enhanced cold chain logistics performance in refrigerated transport. 2024 2nd International Conference on Sustainable Computing and Smart Systems (ICSCSS), Coimbatore, India, 379–384 (2024). https://doi.org/10.1109/ICSCSS60660.2024.10624822
33. A. Giri, A. V. Sagarnal, C. R, A. Dammur, Enhancing efficiency in smart grid billing with blockchain technology and smart contracts. 2024 2nd International Conference on Sustainable Computing and Smart Systems (ICSCSS), Coimbatore, India, 586–594 (2024). https://doi.org/10.1109/ICSCSS60660.2024.10625244
34. M. Sathishkumar and V. Raghavendran, Securing IoT healthcare data: the power of blockchain and homomorphic encryption. 2024 2nd International Conference on Sustainable Computing and Smart Systems (ICSCSS), Coimbatore, India, 309–314 (2024). https://doi.org/10.1109/ICSCSS60660.2024.10624820
35. J. R. Francis Raj, G. R. Sankar, M. Karthihadevi, R. S. Krishnan, C. Karthiga and J. N. Jothi, AI-powered smart waste bins: improving hostel mess cleanliness with IoT and machine learning. 2024 2nd International Conference on Sustainable Computing and Smart Systems (ICSCSS), Coimbatore, India, 452–459 (2024). https://doi.org/10.1109/ICSCSS60660.2024.10625369
36. T. Yagawa, T. Teruya, K. Suzaki and H. Abe, Delegating verification for remote attestation using TEE. 2024 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW), Vienna, Austria, 186–192 (2024). https://doi.org/10.1109/EuroSPW61312.2024.00025
37. A. Hussain et al., Ensuring zero trust IoT data privacy: differential privacy in blockchain using federated learning. In IEEE Transactions on Consumer Electronics. https://doi.org/10.1109/TCE.2024.3444824
38. L. Sun, L. Liu, C. Ma and H. Zhu, Reach on access control technology of geomagnetic observation network system based on fog computing. 2024 2nd International Conference On Mobile Internet, Cloud Computing and Information Security (MICCIS), Changsha City, China, 215–220 (2024). https://doi.org/10.1109/MICCIS63508.2024.00042
39. S. Mehta and D. Kundra, Improving drone operational efficiency through IoT-blockchain integration. 2023 4th International Conference on

Intelligent Technologies (CONIT), Bangalore, India, 1–5 (2024). https://doi.org/10.1109/CONIT61985.2024.10626752
40. I. P. Okokpujie, K. -O. A. Kinggs, A. O. Onokwai, A. Adetunla and C. O. Osueke, Review and analysis of IoT-based control systems in supply chain distribution, home applications, irrigation monitoring, and energy generation: challenges and future trends. 2024 International Conference on Science, Engineering and Business for Driving Sustainable Development Goals (SEB4SDG), Omu-Aran, Nigeria, 1–24 (2024). https://doi.org/10.1109/SEB4SDG60871.2024.10630352
41. O. O. Afolalu, O. A. Akpor, S. A. Afolalu and O. F. Afolalu, Internet of Things applications in health systems' equipment: challenges and trends in the Fourth Industrial Revolution. 2024 International Conference on Science, Engineering and Business for Driving Sustainable Development Goals (SEB4SDG), Omu-Aran, Nigeria, 1–9 (2024). https://doi.org/10.1109/SEB4SDG60871.2024.10629785
42. A.A. Alshdadi, A.A. Almazroi, E. Alsolami, N. Ayub, M.D. Lytras, Enhanced IoT security for DDOS attack detection: split attention-based ResNeXt-GRU ensembler approach. IEEE Access **12**, 112368–112380 (2024). https://doi.org/10.1109/ACCESS.2024.3443067
43. A. V. Pomogalova, E. A. Donskov, V. S. Elagin and A. G. Vladyko, Aspects of data transfer and synchronization for vulnerable road users emergency scenarios based on blockchain technology in ITS. 2024 Systems of Signal Synchronization, Generating and Processing in Telecommunications (SYNCHROINFO), Vyborg, Russian Federation, 1–6 (2024). https://doi.org/10.1109/SYNCHROINFO61835.2024.10617888
44. C. Kamienski, I. Zyrianoff, L. F. Bittencourt and M. Di Felice, IoTinuum: the IoT computing continuum. 2024 20th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), Abu Dhabi, United Arab Emirates, 732–739 (2024). https://doi.org/10.1109/DCOSS-IoT61029.2024.00112
45. R. Mukkamala, S. Olariu, M. Aljohani, M. Sunkara and H. Aldabagh, An adaptive drone-based multigenerational sensor system for monitoring large ecosystems. 2024 20th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), Abu Dhabi, United Arab Emirates, 361–368 (2024). https://doi.org/10.1109/DCOSS-IoT61029.2024.00061
46. R. Trabelsi, G. Fersi and M. Jmaiel, A fog and blockchain-based distributed virtual private networks (VPN). 2024 20th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), Abu Dhabi, United Arab Emirates, 130–137 (2024). https://doi.org/10.1109/DCOSS-IoT61029.2024.00028
47. M. S. I. Noyon, N. Moradpoor, L. Maglaras and J. Ahmad, Enhancing cloud computing security through blockchain-based communication for electronic health records. 2024 20th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), Abu Dhabi, United Arab Emirates, 516–523 (2024). https://doi.org/10.1109/DCOSS-IoT61029.2024.00082
48. G. Andronikidis, G. Niotis, C. Eleftheriadis, K. Kyranou, S. Nikoletseas and P. Sarigiannidis, Optimizing federated learning through lightweight and scalable blockchain. 2024 20th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), Abu Dhabi, United Arab Emirates, 469–476 (2024). https://doi.org/10.1109/DCOSS-IoT61029.2024.00075
49. D. Das, U. Ghosh, N. Evans and S. Shetty, Blockchain-enabled secure device-to-device communication in software-defined networking. 2024 IEEE International Conference on Communications Workshops (ICC Workshops), Denver, CO, USA, 1450–1455 (2024). https://doi.org/10.1109/ICCWorkshops59551.2024.10615611
50. M.M. Asiri, H. Alfraihi, Y. Said, K.M. Othman, A.S. Salama, R. Marzouk, Securing consumer electronics devices: a blockchain-based access management approach enhanced by deep learning threat modeling for IoT ecosystems. IEEE Access **12**, 110671–110680 (2024). https://doi.org/10.1109/ACCESS.2024.3441094
51. A. Heidari, N. Jafari Navimipour, M.A. Jabraeil Jamali et al., Securing and optimizing IoT offloading with blockchain and deep reinforcement learning in multi-user environments. Wireless Netw. **31**, 3255–3276 (2025). https://doi.org/10.1007/s11276-025-03932-4
52. A. Heidari, N.J. Navimipour, M. Unal, A secure intrusion detection platform using blockchain and radial basis function neural networks for Internet of Drones. IEEE Internet of Things J. **10**(10), 8445–8454 (2023). https://doi.org/10.1109/JIOT.2023.3237661
53. A. Heidari, Z. Amiri, M.A.J. Jamali, N. Jafari, Assessment of reliability and availability of wireless sensor networks in industrial applications by considering permanent faults. Concurrency Computat Pract Exper. **36**(27), e8252 (2024). https://doi.org/10.1002/cpe.8252
54. Z. Amiri, A. Heidari, M. Zavvar, N.J. Navimipour, M. Esmaeilpour, The applications of nature-inspired algorithms in Internet of Things-based healthcare service: a systematic literature review. Trans Emerging Tel Tech **35**(6)(2024). https://doi.org/10.1002/ett.4969
55. Z. Amiri, A. Heidari, N.J. Navimipour, Comprehensive survey of artificial intelligence techniques and strategies for climate change mitigation. Energy. 308, 132827 (2024). https://doi.org/10.1016/j.energy.2024.132827
56. S. Toumaj, A. Heidari, R. Shahhosseini et al., Applications of deep learning in Alzheimer's disease: a systematic literature review of current trends, methodologies, challenges, innovations, and future directions. Artif. Intell. Rev. **58**, 44 (2025). https://doi.org/10.1007/s10462-024-11041-5
57. Amiri Z, Heidari A, Jafari N, Hosseinzadeh M. Deep study on autonomous learning techniques for complex pattern recognition in interconnected information systems. Computer Science Review. 2024 Nov 1;54:100666.https://doi.org/10.1016/j.cosrev.2024.100666
58. J. Jamali, M. Ali, et al., The IoT landscape. Towards the Internet of Things: Architectures, Security, and Applications 1–8 (2020). https://doi.org/10.1002/cpe.8349
59. S.S. Ahmadpour, M. Zaker, N.J. Navimipour, N.K. Misra, M. Zohaib, S. V, A. Heidari, A.H. Navin, M. Hosseinzadeh, M. Hakimi, A nano-design of a quantum-based arithmetic and logic unit for enhancing the efficiency of the future IoT applications. AIP Advances **15**(3), 035341 (2025). https://doi.org/10.1063/5.0247642
60. P. Chinnasamy, A. Albakri, M. Khan, A.A. Raja, A. Kiran, J.C. Babu, Smart contract-enabled secure sharing of health data for a mobile cloud-based e-health system. Appl. Sci. **13**(6), 3970 (2023). https://doi.org/10.3390/app13063970
61. P. Chinnasamy, G.C. Babu, R.K. Ayyasamy, S. Amutha, K. Sinha, A. Balaram, Blockchain 6G-based wireless network security management with optimization using machine learning techniques. Sensors **24**(18), 6143 (2024). https://doi.org/10.3390/s24186143
62. R. Suganya1, et al., "Wireless mmWave communication in 5G network slicing with routing model based on IoT and deep learning model", Transactions on Emerging Telecommunications Technologies, 2025; 36:e70071 .https://doi.org/10.1002/ett.70071
63. P. Chinnasamy 1, et al., Defending against digital threats: machine learning techniques for cyber persecution detection. 2024 5th International Conference on Artificial Intelligence and Data Sciences (AiDAS). https://doi.org/10.1109/AiDAS63860.2024.10730176
64. P.Chinnasamy 1, et al., Enhancing healthcare record privacy through the integration of visual cryptography and diverse image encryption techniques. Proceedings of the Nineth International Conference on Communication and Electronics Systems (ICCES-2024) IEEE Xplore Part Number: CFP24AWO-ART; ISBN: 979–8–3503–7797–2 https://doi.org/10.1109/ICCES63552.2024.10859628
65. K Zanbouri, et al. ,"A GSO-based multi-objective technique for performance optimization of blockchain-based industrial Internet of Things", Int J Commun Syst. 2024;37:e5886. wileyonlinelibrary.com/journal/dac © 2024 John Wiley & Sons Ltd. https://doi.org/10.1002/dac.5886
66. M. Asadi1, et al., Botnets unveiled: a comprehensive survey on evolving threats and defense strategies. Transactions on Emerging Telecommunications Technologies, https://doi.org/10.1002/ett.5056.
67. A. Ayub Khan, A. A. Laghari, Z. A. Shaikh, Z. Dacko-Pikiewicz and S. Kot, Internet of Things (IoT) security with blockchain technology: a state-of-the-art review. In IEEE Access, 10, 122679–122695 (2022). https://doi.org/10.1109/ACCESS.2022.3223370

Publisher's Note